

TI DSPs HARDWARE and BLOCK CODING CONCEPTS with Simulink or PSIM

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Content

First hour

1. TI DSP Family
2. C2000 Series - DRV8312EVM

Second Hour

3. Applications (Insta-Spin – Insta-Motion)
4. Matlab/Simulink and PSIM Interface

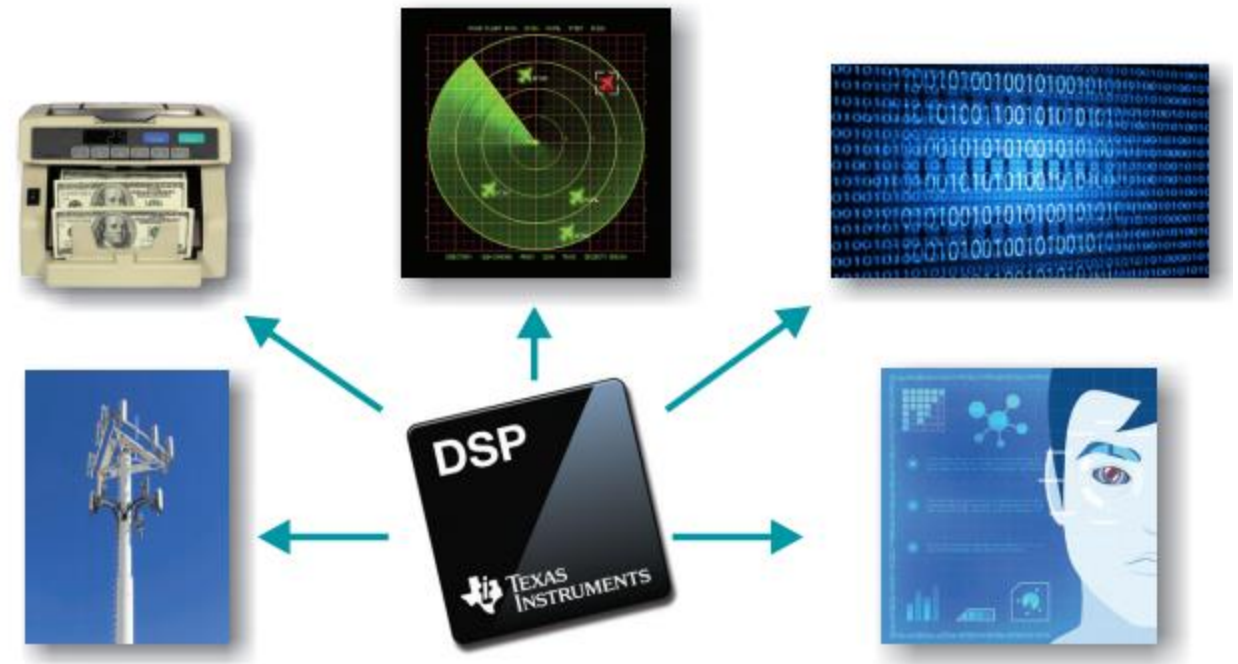
DSP (Overview)

Digital Signal Processor (DSP)

- small
- low power
- smart

Multi-core compute platform

- ✓ Voice & image recognition
- ✓ Real-time data analytics
- ✓ Control
- ... endless



TI DSP Family - Applications

	MSP Microcontrollers	C2000™ Microcontrollers	Sitara™ ARM® Cortex®-A Processors	AWR and IWR mmWave Sensors	C5000™ / C6000™ Digital Signal Processors	Jacinto™ Automotive Processors	SimpleLink™ Wireless Microcontrollers
Audio							
Automotive							
Communications							
Industrial							
Medical							
Security							
Video							
Wireless							
Key feature	Ultra-low power	Real-time control	Scalable performance with robust peripheral options	High-accuracy single-chip sensor with processor	Scalable performance from ultra-low power to high performance	ADAS and infotainment systems	Easy-to-use, low-power connectivity

TI – DSP Family : Low Power MCUs

- MSP 430

- Offer the **lowest power consumption** and the perfect mix of integrated peripherals with a common ecosystem for a **wide range of sensing and measurement applications**.

- SimpleLink

- Offers the broadest portfolio of **32-bit ARM-based micro-controllers** with **industry-leading** features including low power, robustness and integrated security to support **more than ten differentiated wired and wireless protocols including Bluetooth® low energy, Sub-1 GHz and Wi-Fi®**. Each device offers IoT developers a number of features to uniquely solve their problems, whether capturing **high-precision 16-bit analog signal, enabling more security**.

Low-Power MCUs MSP430™ Microcontrollers

Ultra-Low-Power, 16-Bit Microcontrollers for Sensing and Measurement Applications

Key Applications

- Intelligent sensors and control
- Consumer electronics
- Test & measurement
- Utility metering
- Portable medical and instrumentation
- Security systems

Series	Capacitive-Sensing MCUs	Value-Line Sensing MCUs		Performance-Sensing MCUs		Other MSP430 MCUs	
Family	MSP430FR25x/ FR26x	MSP430FR2xx/ FR4x	MSP430G2x/ i2x	MSP430FR5x/ FR6x	MSP430F5x/ F6x	MSP430F2x/ F4x	MSP430F1x
Max speed (MHz)	16	16	16	16	25	16	16
FRAM (max KB)	16	16	56	256	512	120	120
SRAM (max KB)	4	4	4	8	67	8	10
GPIOs	15–19	16–64	4–32	17–83	29–90	14–80	10–48
ADC	•	•	•	•	•	•	•
DAC					•	•	•
Comparator		•	•	•	•	•	•
Timer	•	•	•	•	•	•	•
UART	•	•	•	•	•	•	•
I ² C	•	•	•	•	•	•	•
SPI	•	•	•	•	•	•	•
Multiplier	•	•	•	•	•	•	•
DMA				•	•	•	•
Op Amp		•			•	•	•
LCD		•		•	•	•	
RTC	•	•		•	•	•	
1.8-V I/O					•		
CRC	•	•		•	•		
USB					•		
AES				•	•		

Low-Power MCUs MSP430™ Microcontrollers

Ultra-Low-Power, 16-Bit Microcontrollers for Sensing and Measurement Applications

Key Features

- Ultra-low-power (ULP) architecture enables battery life > 20 years
- 100 μ A/MHz active mode
- 700 nA RAM retention
- 450 nA RTC mode
- 7 low-power modes
- Instantaneous wakeup

Integrated intelligent peripherals to add analog functions and advanced functions including:

- Capacitive touch
- Ultrasonic sensing
- Low-energy vector math engine
- 8- to 12-bit ADC
- Transimpedance amplifier (TIA)
- DMA (Direct Memory Access)
- AES256 (Advanced Encryption Standard)

SimpleLink

Solutions Low-Power Wireless Connectivity

Key Benefits

- Single software foundation
 - SDK based on common foundation of drivers, frameworks and libraries
 - Pre-integrated TI-RTOS kernel already deployed in thousands of products across multiple applications
 - POSIX (IEEE Standard)-compliant API (Application Programming Interface) ensures compatibility with numerous third-party software components

Product	MSP432™	CC3120	CC3220	CC2640R2F	CC1350	CC1310
MCU type	Host MCU	Network processor	Wireless MCU	Wireless MCU	Wireless MCU	Wireless MCU
Application	✓	—	✓	✓	✓	✓
Wireless stack + RF	—	✓	✓	✓	✓	✓
Wireless technology	Connectivity with SDK plug-ins	Wi-Fi®	Wi-Fi	Bluetooth® low energy	Sub-1 GHz + Bluetooth low energy	Sub-1 GHz
Key differentiation	Capture analog signals at up to 16 ENOB using ADC14	Network processor with integrated all Wi-Fi and Internet protocols	Wi-Fi CERTIFIED™ single-chip MCU with enhanced security	Lowest power BT4.2 and BT5 Flash-based solution, auto qualified	World's first dual-band wireless MCU	Low power and longest range to achieve 20 kms on a coin cell
SimpleLink SDK compatible	✓	✓	✓	✓	✓	✓

SimpleLink Portfolio Development Kits

MSP-EXP432P401R	CC3220SF-LAUNCHXL	LAUNCHXL-CC2640R2
• SimpleLink MSP432 LaunchPad™ kit • Integrated 14-bit ADC • Host MCU	• SimpleLink CC3220 LaunchPad kit • Wi-Fi support • Wireless MCU	• SimpleLink CC2640R2F LaunchPad kit • Bluetooth low energy support • Wireless MCU
LAUNCHXL-CC1350	LAUNCHXL-CC1310	CC3120BOOST
• SimpleLink CC1350 LaunchPad kit • Sub-1 GHz + Bluetooth low energy support • Wireless MCU	• SimpleLink CC1310 LaunchPad kit • Sub-1 GHz support • Wireless MCU	• SimpleLink CC1350 BoosterPack™ plug-in module • Sub-1 GHz + Bluetooth low energy support • Wireless MCU

TI – DSP Family : Performance MCUs

- C2000
 - Real-Time Control
 - 32-bit MCUs are based on the industry-leading C28x CPU. This CPU features a powerful signal processing engine capable of addressing demanding closed loop tasks thereby giving designers the means to improve system efficiency, reliability and flexibility.

Performance MCUs

C2000™ Real-Time Control Microcontrollers

Performance MCUs for Applications Needing Low-Latency Closed-Loop Control

Applications

- Digital motor control (white goods, industrial drives, medical)
- Digital power supplies (telecom and serv-er rectifiers, wireless base-stations, UPS)
- Renewable energy (solar, wind, fuel cells)
- Intelligent LED lighting
- Automotive (HEV/EV, electric power steering, driver's assistance radar, wip-ers, HVAC, pumps)
- Power line communications (PLC)

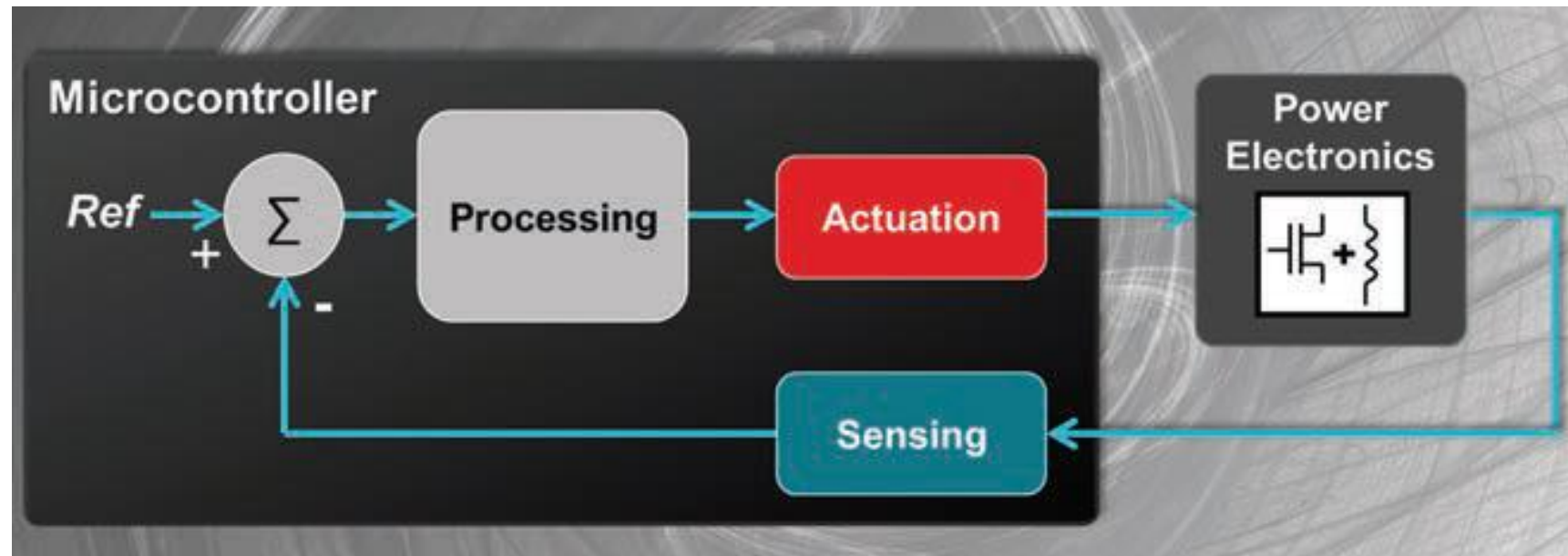
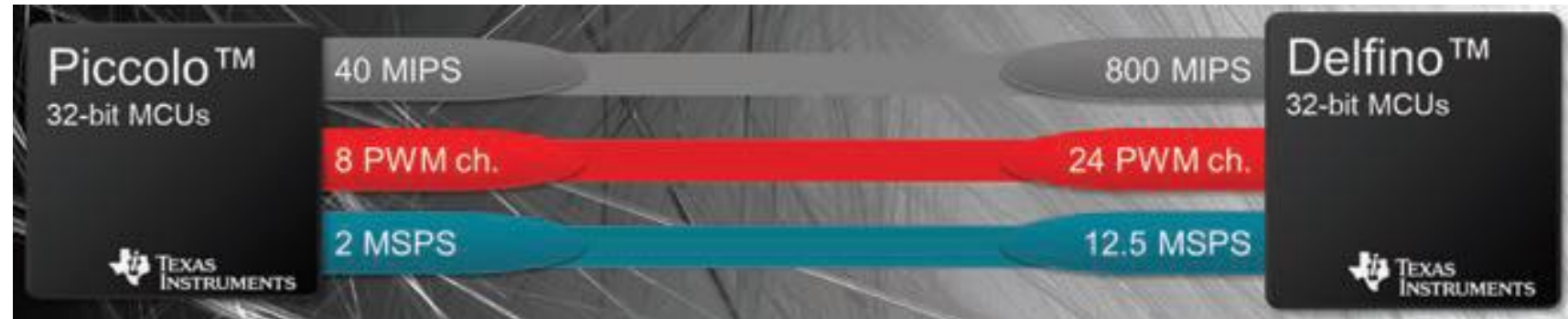
Performance MCUs

C2000™ Real-Time Control Microcontrollers

Performance MCUs for Applications Needing Low-Latency Closed-Loop Control

Built for Real-Time
Control Optimized
architecture for

- **processing,**
 - **sensing,**
 - **actuation**
- to increase closed-loop performance.



Performance MCUs

C2000™ Real-Time Control Microcontrollers

Performance MCUs for Applications Needing Low-Latency Closed-Loop Control

Processing

- More performance **per MHz with 32-bit C28x DSP** core optimized for complex single-cycle operations common to control theory
- Meet the demands of a wide range of applications with optimized processing options from **40 MIPS to 800 MIPS** of performance
- Add functionality with the Control Law Accelerator (CLA) processing engine. Great for **controlling multiple motors, power stages and more**
- Accelerate complex control theory and signal processing, such as **trigonometric math, FFTs, and complex math**, with built-in hardware accelerators

MIPS : Million instructions per second

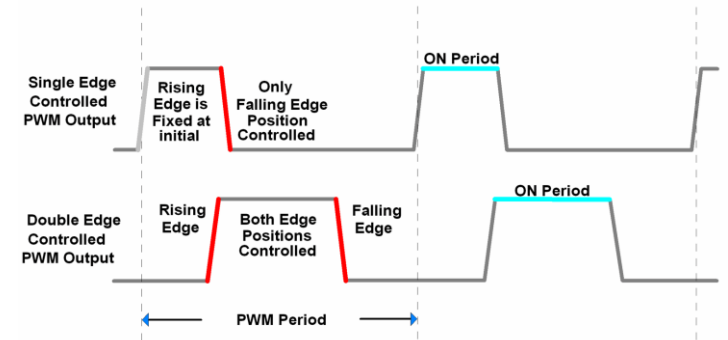
Performance MCUs

C2000™ Real-Time Control Microcontrollers

Performance MCUs for Applications Needing Low-Latency Closed-Loop Control

Actuation

- Achieve higher system performance with **micro edge positioning of PWM outputs**, including support of **PWM phase, duty cycle and period**
- Control a variety of applications and power-stage topologies with ultra-configurable PWM generation
- Support even **complex peak current mode control** of power stages through ramp-generation logic and integrated analog comparators



Performance MCUs

C2000™ Real-Time Control Microcontrollers

Performance MCUs for Applications Needing Low-Latency Closed-Loop Control

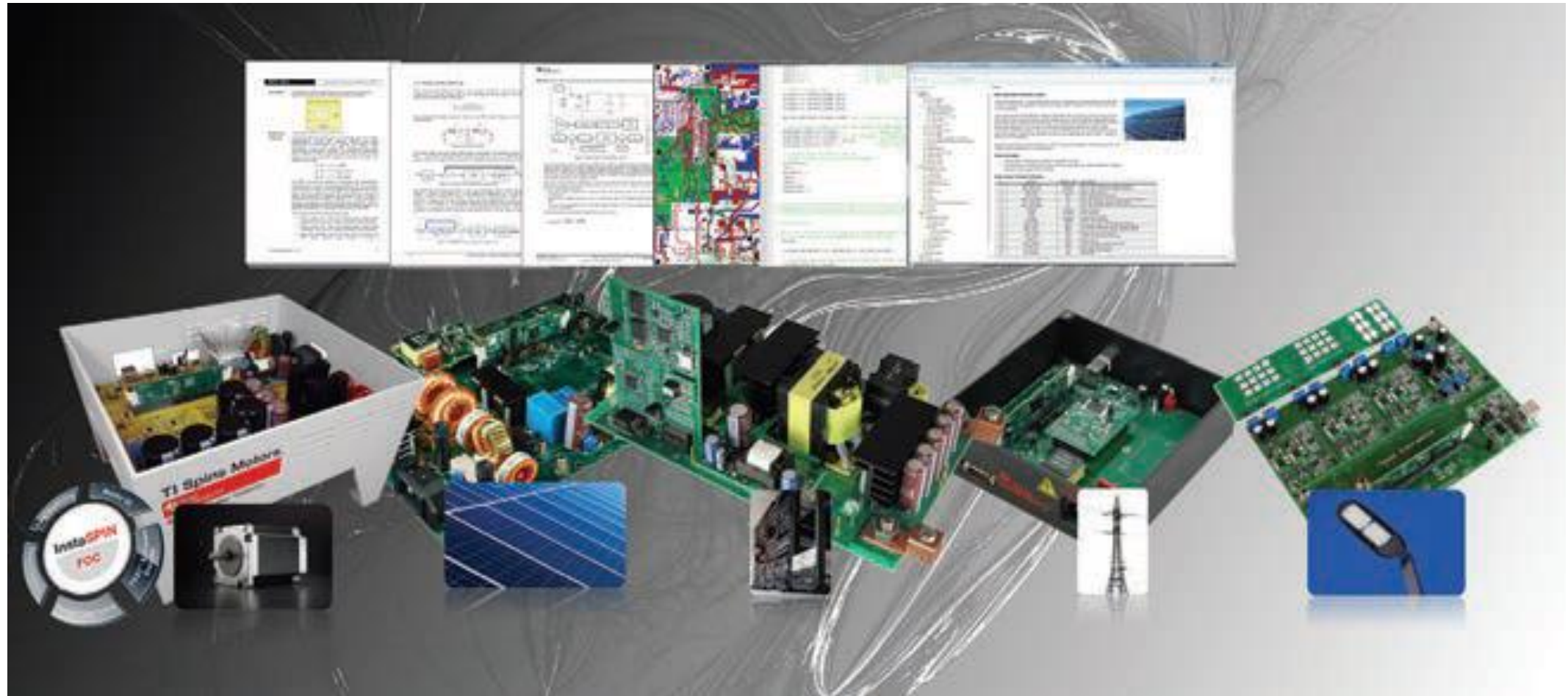
Sensing

- Accurately sample signals with **12-bit and 16-bit** analog-to-digital converters (**ADC**)
- Run systems at high frequencies with ADC conversion rates up to **12.5 MSPS**
- Protect systems better with responsive analog comparators – **30 ns response time**
- Interface with **high-performance external sensors** using C2000 high resolution captures

Performance MCUs

C2000™ Real-Time Control Microcontrollers

Performance MCUs for Applications Needing Low-Latency Closed-Loop Control



<http://www.ti.com/microcontrollers/c2000-real-time-control-mcus/design-development.html>

TI – DSP Family : Sitara Processors

- Sitara
 - TI's Sitara processors are optimized, scalable solutions with a broad portfolio of **ARM Cortex-A8, Cortex-A9 and Cortex-A15** solutions. Sitara processors help innovators create a variety of **feature-rich, low-power applications**. These devices provide robust peripheral support including **3D graphics** and **industrial communications protocols** among others, giving customers flexibility in design. Bringing together flexible hard-ware solutions with robust software tools enables customers using **Sitara** processors to create inspiring ARM designs that start to **bridge the gap between high-end MCUs and low-end ARM processor-based devices**.

Sitara Processors

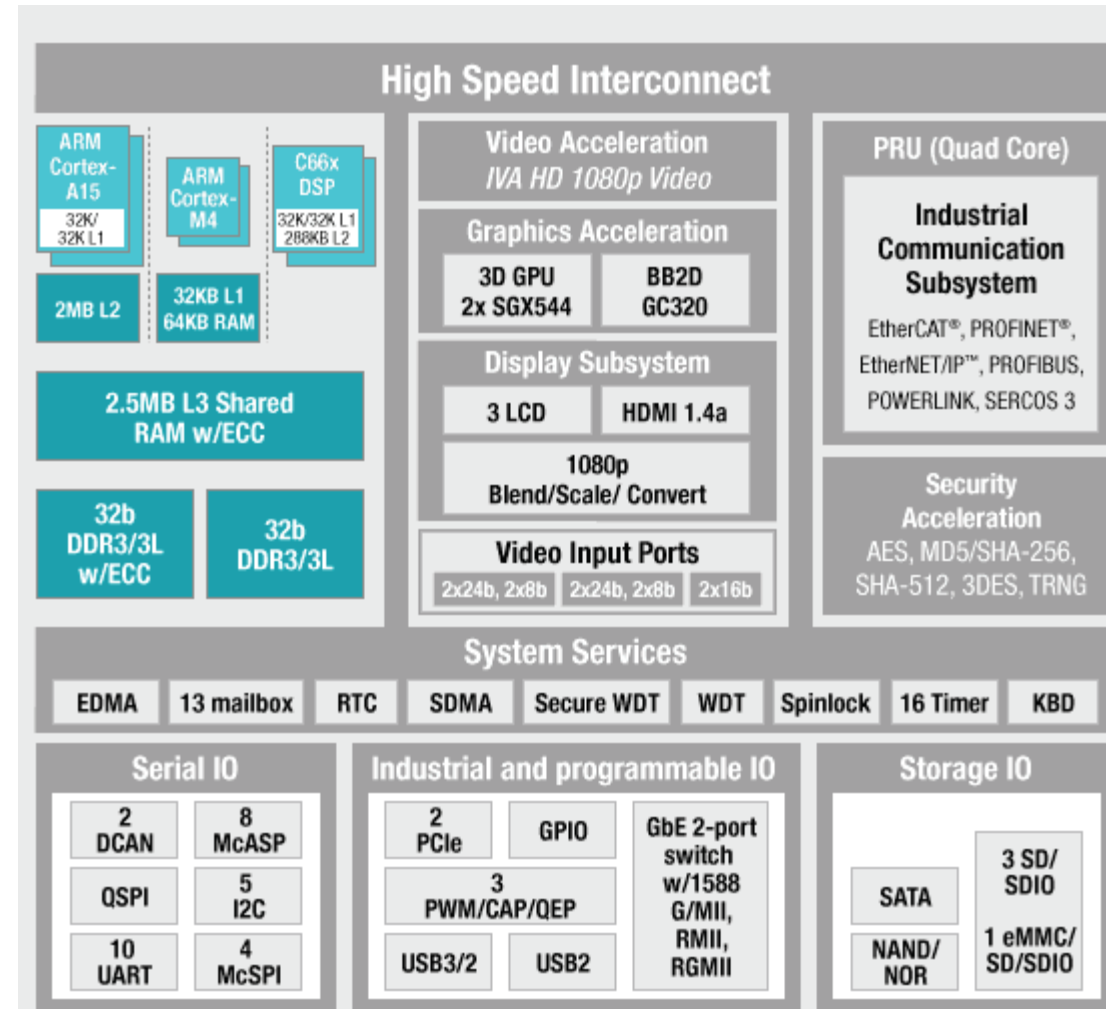
Highly Integrated; Scalable From 300 MHz to 1.5 GHz with 3D Graphics

Specifications

- ARM® Cortex®-A8 and Cortex-A15 processors
- Optional high-performance 3D graphics accelerator
- Optional Programmable Real-time Unit and Industrial Communication Sub-System (PRU-ICSS)
- Robust display options
- Optimized for low-power applications
- Linux®, Android™, RTOS and StarterWare™ software

Targeted Applications

- Factory automation and control, human machine interface, medical, portable data terminals



TI – DSP Family : C5000-C6000

- C5000, C6000
 - The ultra-low-power C5000 fixed-point DSPs offer processors with **active core power below 0.15 mW/MHz**, great for **battery-powered applications**. The C6000 family features fixed and floating point DSPs starting at **200 MHz** and going as high as **1.25 GHz**. The KeyStone multicore family of products use our **highest performance floating point C66x DSP core** as well as **ARM Cortex-A15** CPUs in high-performance system-on-chips (SoCs) that deliver nearly **200 GFLOPS** of performance for less than 20 W. Whether you are developing a battery powered audio processing system, a higher-performance signal processing machine or systems in between, the breadth of the TI DSP processor portfolio provides a likely alternative for your project.

C5000, C6000

Ultra-Low Power, Floating-Point and High-Performance Signal Processing

Targeted Applications

- AUDIO A/V receivers and soundbars Professional audio equipment Voice assistant and user interface
- RADIO AND COMMUNICATIONS Land mobile radio First responder communications
- DEFENSE AND AEROSPACE Radar systems Communications systems Flight control systems
- GRID INFRASTRUCTURE Power quality monitoring Signal analysis Grid communications
- MACHINE VISION Optical inspection equipment Robotic guidance systems Process control

TI – DSP Family : Jantino Automotive Processors

- Jacinto

- From entry-level to premium system-on-chips (SoCs), TI offers a wide range of innovative auto-motive processors for the modern automobile. With a scalable architecture across Infotainment and ADAS solutions, TI provides maximum optimization for automotive OEMs and Tier 1 customers.

- ADAS applications:

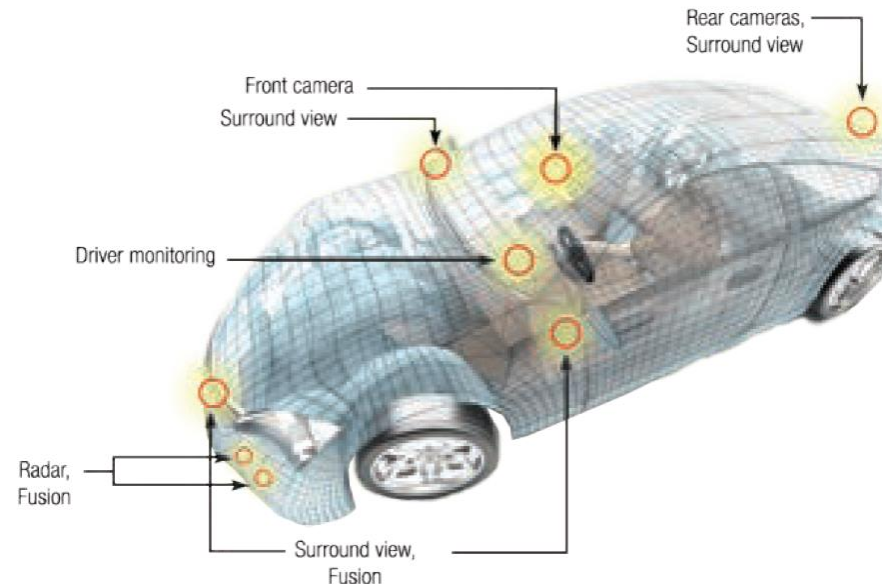
Front camera

Surround view/rear camera

Fusion

Radar

Driver monitoring



Development Kits, Software and Tools

Comprehensive, Easy-to-Use Development Kits, EVMs, Software and Tools

Development kits &
Evaluation modules:

TI embedded family	Featured development kits
MSP430™ MCUs	MSP-EXP430FR5994 LaunchPad™ kit and MSP-EXP430FR4133 LaunchPad kit. See ti.com/msp for complete listing.
SimpleLink™ MCUs	SimpleLink LaunchPad development kits: MSP-EXP432P401R , CC3220SF-LAUNCHXL , LAUNCHXL-CC2640R2 , LAUNCHXL-CC1350 , and LAUNCHXL-CC1310 . See ti.com/simplelink
C2000™ MCUs	Delfino™ LaunchPad Kits: F28379D , F28377S Piccolo™ LaunchPad Kits: F28027 , InstaSPIN-MOTION See ti.com/c2000 for a complete listing
Sitara™, DSP and multicore processors	Starter kits for low-cost application development: TMDSSK3358 , TMDXSK437x Evaluation module for high-performance Sitara processors: TMDSEVM572X Evaluation module for the C66x DSP: TMDSEVM6678
mmWave sensors	Evaluation modules for the mmWave sensor portfolio: IWR1443BOOST , IWR1642BOOST , AWR1243BOOST , AWR1443BOOST and AWR1642BOOST
Automotive processors	TDax ADAS tools and EVMs and DRax Infotainment tools and EVMs

Development Kits, Software and Tools

Comprehensive, Easy-to-Use Development Kits, EVMs, Software and Tools

Software:

TI embedded family	Primary software
MSP430 MCUs	MSP430Ware is a collection of resources to create and build MSP430 MCU code. MSPWare is available as a component of Code Composer Studio™ IDE desktop and cloud versions, or as a standalone package.
SimpleLink MCUs	The SimpleLink SDK is a single software development environment for SimpleLink MCUs that offers flexible software and tool options, 100 percent code reuse and unprecedented scalability for IoT applications.
C2000 MCUs	controlSUITE™ for C2000™ microcontrollers is a cohesive set of software infrastructure, tools and documentation designed to minimize system development time. Further simplifying design efforts, powerSUITE for digital power designers and DesignDRIVE for industrial drive designers is included.
Sitara, DSP and multicore processors	Processor SDK is a unified software platform for TI's processors providing easy setup and fast out-of-the-box access to benchmarks and demos for Linux®, RT-Linux, TI-RTOS and Android™ operating systems.
mmWave sensors	mmWave SDK is a software platform for the TI mmWave Sensing Portfolio, providing the building blocks, demonstrations/examples, tools and documentation to begin development.
Automotive processors	Processor SDK Automotive is the foundational software development platform for TI's Jacinto DRAX and TDAX family of Infotainment and ADAS SoCs.

Development Kits, Software and Tools

Comprehensive, Easy-to-Use Development Kits, EVMs, Software and Tools

Development
Tools:

TI embedded family	Featured development tools
MSP430 MCUs	MSP-FET debugger, EnergyTrace™ energy-based code analysis tool
SimpleLink MCUs	SimpleLink Academy training modules, TI Resource Explorer
C2000 MCUs	XDS110 Debug probe, UniFlash Flash programmer
Sitara, DSP and multicore processors	Clock tree tool, Pin MUX tool, Power estimation tool
mmWave sensors	mmWave sensing estimator, mmWave demo visualizer

Development Kits, Software and Tools

Comprehensive, Easy-to-Use Development Kits, EVMs, Software and Tools

Code Composer Studio™ (CCS)

is a free integrated development environment (IDE) that supports TI's Microcontroller and Embedded Processors portfolios. CCS IDE comprises a suite of tools used to develop and debug embedded applications.

CCS includes

- C/C++ compiler
- Source code editor
- Project build environment
- Debugger
- Profiler

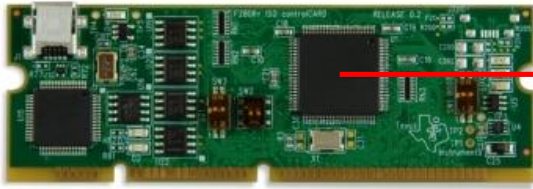
TI also supports **Energia**, an open-source and community-driven Integrated Development Environment (IDE) and software framework with its microcontroller portfolio.

TI C2000 Series – DRV8312EVM



BLDC/PMSM MOTOR - DRIVER BOARD - DSP BOARD

TI C2000 Series DRV8312EVM

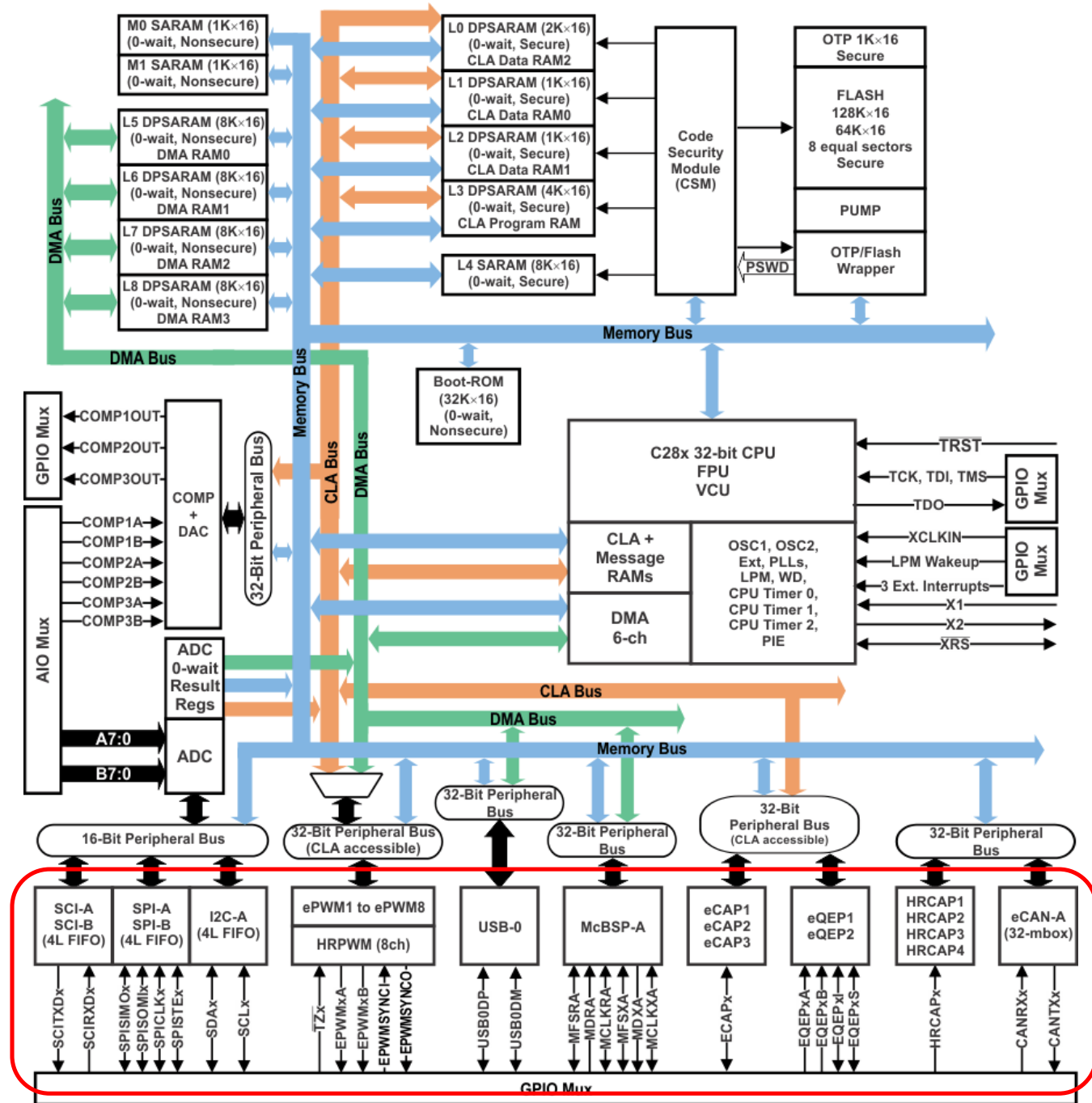


- DSP BOARD

Piccolo - TMDSCNCD28069MISO
controlCARD

- DSP

TMS320F28069M, Piccolo™ 32-bit
MCU with 90 MHz, FPU, VCU, 256
KB Flash, CLA, InstaSPIN-MOTION



TI C2000 Series DRV8312EVM

- DRV8312

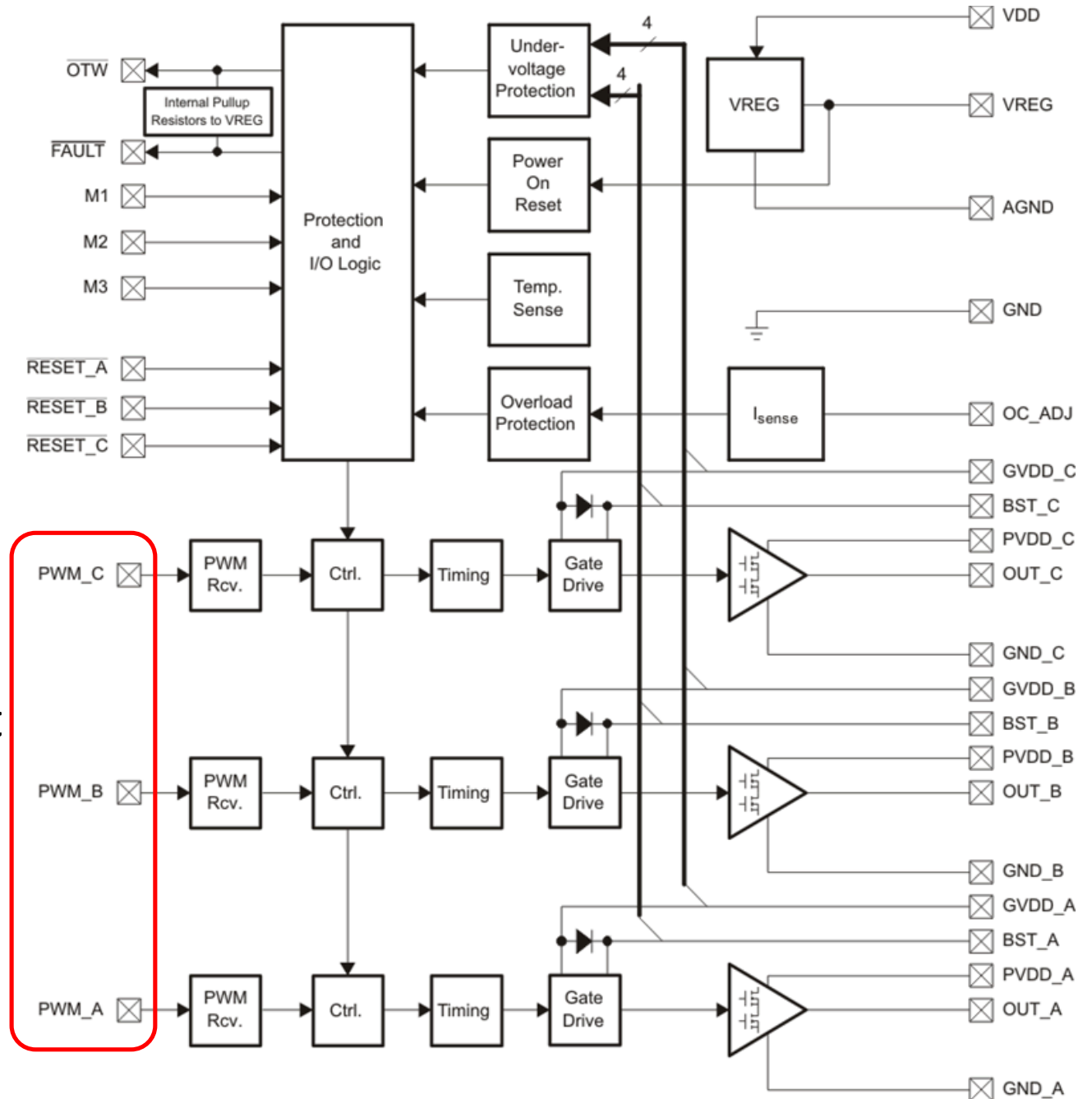
6.5A three phase brushless DC motor driver with inrush protection (PWM ctrl)

PWM Operating Frequency up to 500 kHz.

Integrated Self-Protection Circuits Including Undervoltage, Overtemperature, Overload, and Short Circuit.

High-Efficiency Power Stage (up to 97%) With Low RDS(on) MOSFETs (80 mΩ at T_J = 25°C)

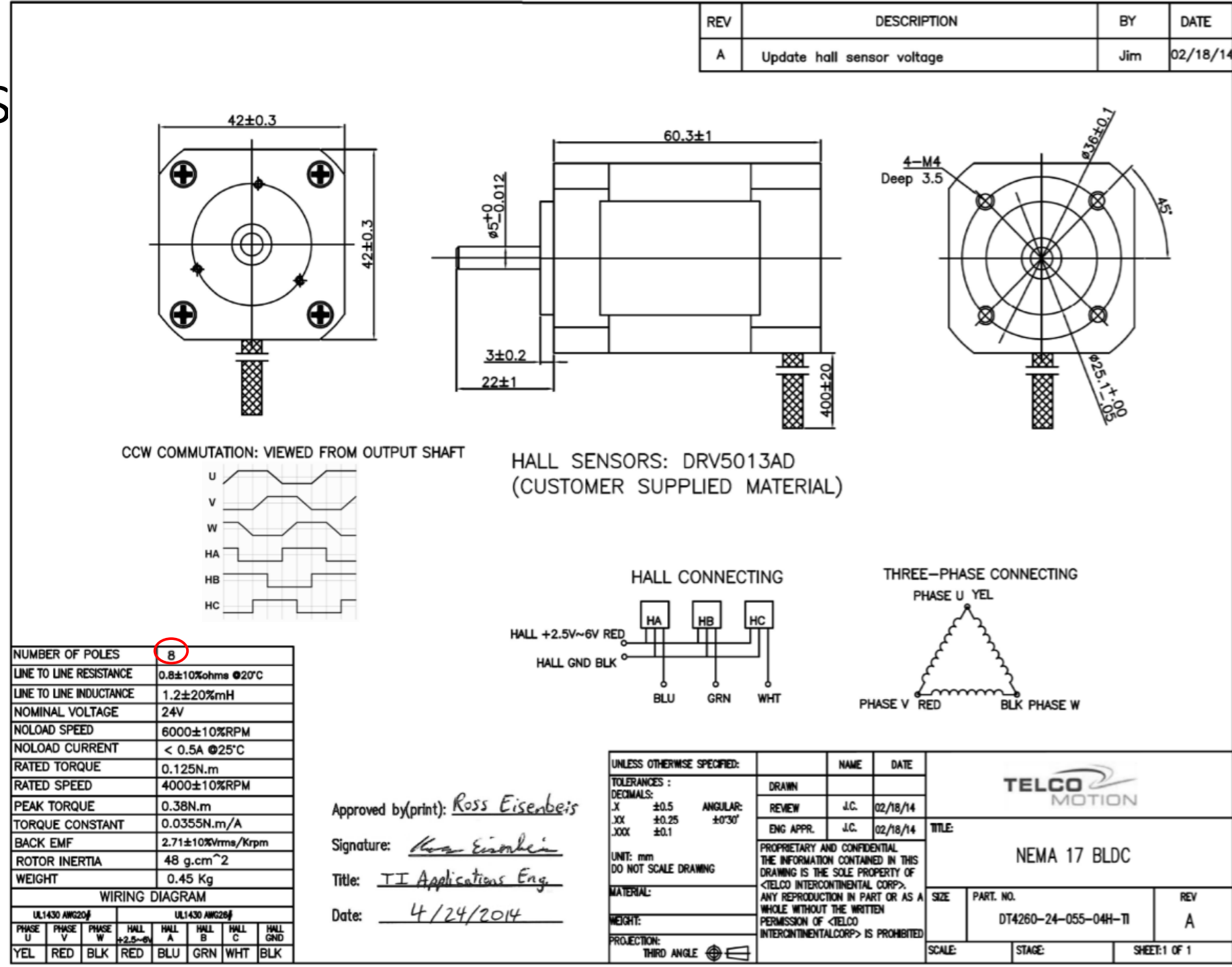
Operating Supply Voltage up to 50 V (70-A Absolute Maximum)



TI C2000 Series DRV8312EVM

- MOTOR (BLDC)

DT4260-24-055-
04H-PT 24VDC
4000RPM 1814-
0067MOTOR
(telcointercon)



InstaSPIN - Motion



InstaSPIN-MOTION features

Identify inertia

- Uses inertia to provide the most accurate system control across a wide range of inertia change
- The SpinTAC inertia estimator automatically measures system inertia by spinning the motor and measuring feedback

Simple tuning

- Single coefficient tuning to quickly test and tune your control from soft to stiff response
- Reduce complexity and system tuning time in multi-variable PID based systems with single gain (bandwidth) that works across the entire variable speed, position and load range of an application
- A single parameter tunes both position and speed

Disturbance rejection

- SpinTAC controller automatically rejects disturbances caused by:
 - cycle transitions
 - changing loads
 - environmental disturbances
- Maximize control and recover more quickly than a PI controller, with less oscillation, and minimal over/undershoot

Execute motion profiles

- SpinTAC runs on the processor to create smooth configurable trajectories between two positions/speeds
- The Motion Engine automatically generates the best curve to satisfy the user-entered jerk and acceleration constraints for each move

Plan trajectories

- Quickly build various states of motion (point A to point B) and tie them together with state based logic (hold for time, wait for interrupt, move to state C or D based on logic, etc.) in minutes
- Directly embed the trajectories into the C code on the microcontroller

Software architectures

Example projects are designed in a modular structure to demonstrate the flexibility of adding InstaSPIN-MOTION features to your application. Architectures include:

- Sensorless speed control
- Sensored speed control
- Sensored position (servo) control

1.1.3 InstaSPIN-FOC Block Diagrams

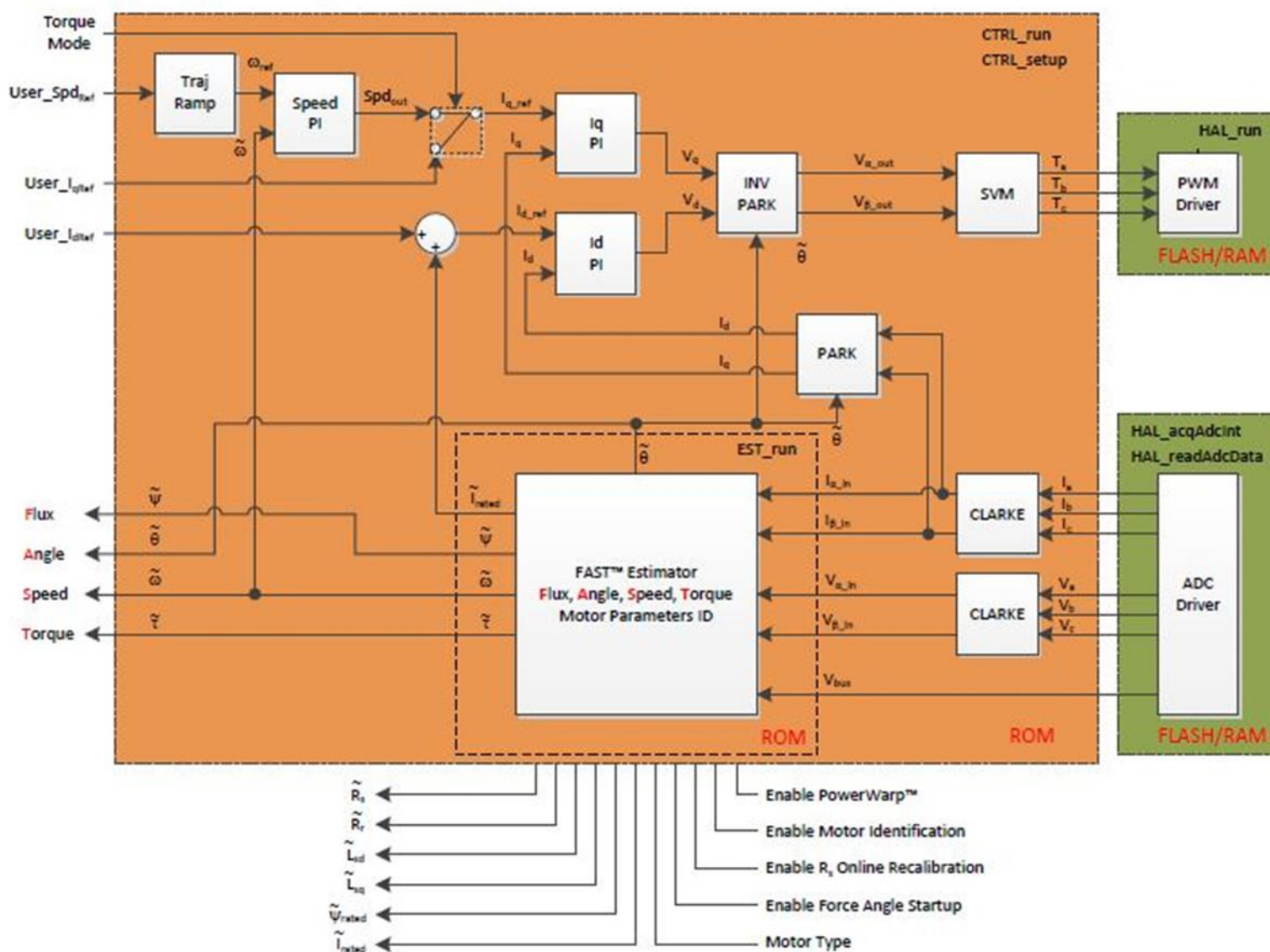


Figure 1-2. i. Block Diagram of Entire InstaSPIN-FOC Package in ROM (except F2802xF devices)

InstaSPIN - Motion

InstaSPIN_MOTION_F2806xM

File

Configuration Application

Motor Identification SpinTAC 1:Startup SpinTAC 2:Tuning SpinTAC 3:Motion Speed or Torque Speed Profiles Advanced: Field Control & Modulation

InstaSPIN™- MOTION

Motor Type

☐ ACIM

☒ PMSM/BLDC

Identification Method

Automatic ID

Identification Settings

Res Est Current(A) 1

Ind Est Current (A) -1

Estimation Freq (Hz) 20

Motor Pole Pairs 4

Motor Max Current (A) 5

Motor Max Freq (Hz) 800

Rated Flux (Vline-line/Hz) 0.0327964

Identification of Motor Parameters and Tuning of Current Loop Controllers

0%

Identified Hardware Target

HV KIT REV1.1

DRV8301 KIT REVD

DRV8312 KIT REVD

HARDWARE NOT IDENTIFIED

FAULT STATUS

Reset DRV Fault

Control Timing

Set in user.h pre-compile

PWM Frequency 20 KHz

ADC Sampling 10 KHz

Current Control 10 KHz

FAST™ 10 KHz

Identify Motor

Status EST_State_Idle

Motor Not Identified

Identifying Motor

Lock Rotor then press Proceed ACIM Only

Proceed

Reset System

Hardware Parameters

ADC Max Volt (V) 66.32

Vbus (V) 24

ADC Max Current (A) 17.3

Voltage Filter Pole (Hz) 714.14

Identified Motor Parameters

R_s (Ω)

L_s_d (H) Identifies average L_s and uses for both d and q. If saliency is known set in user.h

L_s_q (H)

Flux (V/Hz)

Magn Current (A) ACIM Only

R_r (Ω) ACIM Only

InstaSPIN_MOTION_F2806xM

File

Configuration Application

Motor Identification SpinTAC 1:Startup SpinTAC 2:Tuning SpinTAC 3:Motion Speed or Torque Speed Profiles Advanced: Field Control & Modulation

InstaSPIN™- MOTION

Motor Type

☐ ACIM

☒ PMSM/BLDC

Identification Method

Automatic ID

Identification Settings

Res Est Current(A) 1

Ind Est Current (A) -1

Estimation Freq (Hz) 20

Motor Pole Pairs 4

Motor Max Current (A) 5

Motor Max Freq (Hz) 800

Rated Flux (Vline-line/Hz) 0.0327964

Identification of Motor Parameters and Tuning of Current Loop Controllers

100%

Identified Hardware Target

HV KIT REV1.1

DRV8301 KIT REVD

DRV8312 KIT REVD

HARDWARE NOT IDENTIFIED

FAULT STATUS

Reset DRV Fault

Control Timing

Set in user.h pre-compile

PWM Frequency 20 KHz

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Current Control 10 KHz

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Identify Motor

Status EST_State_Idle

Motor Not Identified

Identifying Motor

Lock Rotor then press Proceed ACIM Only

Proceed

Reset System

Hardware Parameters

ADC Max Volt (V) 66.32

Vbus (V) 24

ADC Max Current (A) 17.3

Voltage Filter Pole (Hz) 714.14

Identified Motor Parameters

R_s (Ω) 0.3927605

L_s_d (H) Identifies average L_s and uses for both d and q. If saliency is known set in user.h 0.0006191849

L_s_q (H) 0.0006191849

Flux (V/Hz) 0.0353822

Magn Current (A) ACIM Only

R_r (Ω) ACIM Only



Version: 2.2.1

SpinTAC is a

High-Performance Controller

that replaces PI control

It uses advanced control techniques and is

Tuned with one parameter

In a matter of minutes

You only have to tune it once and
SpinTAC performs across a**Wide Operating Range**

To begin, *check the limits and click "Estimate Motor Inertia"*. After SpinTAC is Enabled, move the tab "SpinTAC 2: Tuning".

1

Limits**Motor Ready**

Ramp Time: 10 [sec]

 $0.1 \leq 10 \leq 25.0$

Iq Limit: 4 [A]

 $0 < 4 \leq 5$

Speed: 2400 [RPM]

 $0 < 2400 \leq 12000$

2

Run[Estimate Motor Inertia](#)

Your motor will spin a few times as SpinTAC estimates the motor inertia

Status**Zero Speed****Identifying****Not Identified****Identified****Results**

Inertia: 0.023306

 $10^{-3} [\text{A}\cdot\text{s}/\text{RPM}]$

Friction: 0.086775

 $10^{-3} [\text{A}\cdot\text{s}/\text{RPM}]$ **SpinTAC Enabled**



Version: 2.2.1



Motor Identified



Motor Running

Start/Stop**Disable SpinTAC**Reverts to Speed PI and
Ramp Reference on
"Speed or Torque" tab

One Minute Tuning

1 Set the Target Speed to zero.**2** Increase the bandwidth until you are satisfied with the tightness or the motor becomes unstable**3** Try a few speeds to make sure the bandwidth settings works across your operating range

Target Speed: 0 [RPM]

Motor Speed: 0 [RPM]

Bandwidth Tuning



[rad/s]

SpinTAC
Speed Controller

Enable



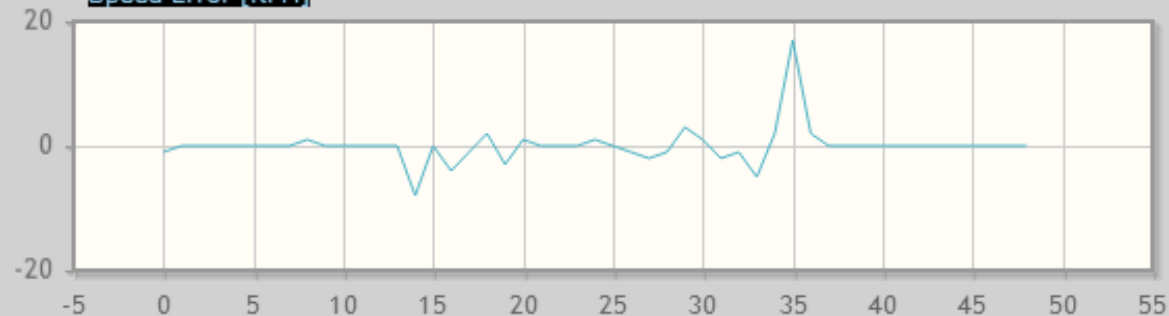
Enable

PI Speed
Controller

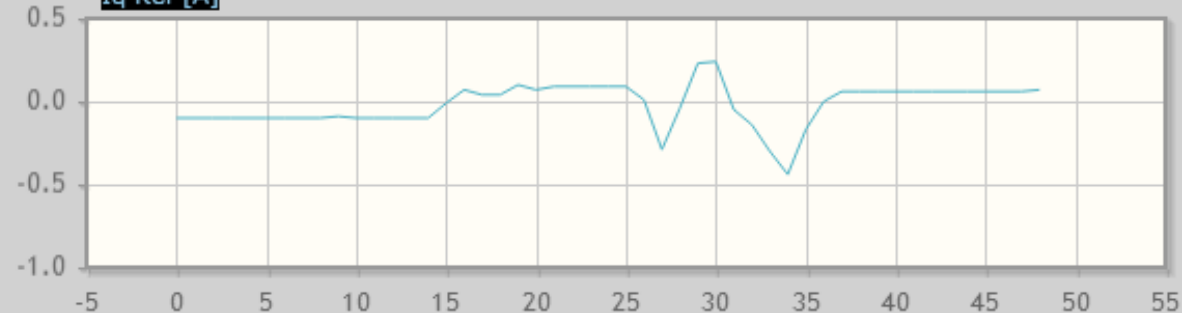
Kp: 8

Ki: 0.08

Speed Error [RPM]



Iq Ref [A]





Version: 2.2.1

Motion Profiler

Choose a curve type and
acceleration/jerk limits

Set a target speed
and hit Enter

Motion profiles will automatically
be generated and sent to the
controller

Note: To use the SpinTAC Profile Generator with a standard PI Controller, click "Enable" on the PI Speed Controller below the graphs

Profile Types

- ☐ Trapezoidal Curve
☐ sCurve
☒ stCurve (Smoothest) *Linestream Proprietary

Profile Limits

Acceleration Limit:

30 ≤ 1200 ≤ 12000 [RPM/s]

Jerk Limit:

0.03 ≤ 2.40 ≤ 192.00 [KRPM/s²]

**Start
Speed**

1000

[RPM]

**Motor
Speed**

1001

[RPM]

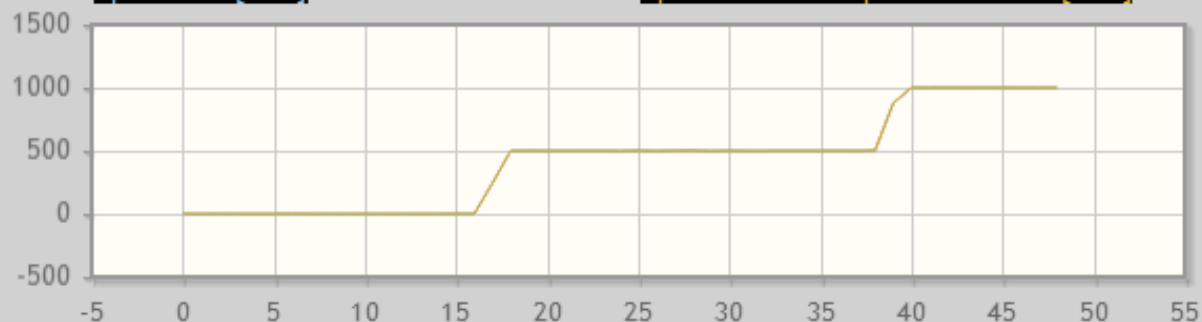
**Target
Speed**

1000

[RPM]

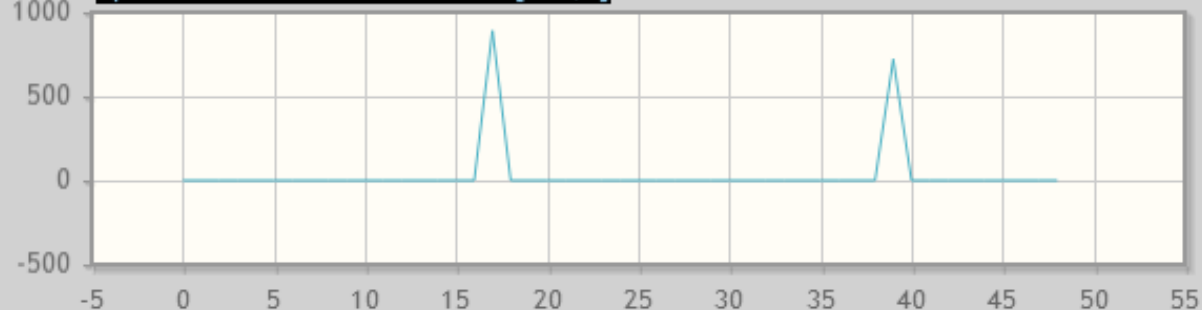
Profile Time: 0 [Sample Counts]

Speed-EST [RPM]



SpinTAC Profiler Speed Reference [RPM]

Spin TAC Profiler Acceleration Ref [RPM/s]



**SpinTAC Profiler
+ SpinTAC Speed
Controller**

Enable



Enable



**SpinTAC Profiler
+ PI Speed Controller**



Motor Identified



Motor Running

Start/Stop

Disable SpinTAC

Reverts to Speed PI and
Ramp Reference on the
"Speed or Torque" tab

Speed Mode

Speed Reference (RPM)

-20000 -10000 0 10000 20000

Max Acceleration (RPM/s)

0 1600 3200 4800 6400 8000

Speed Loop Tuning

To Enable SpinTAC, go to "SpinTAC 1" tab.
Switching is allowed only in "Speed+Torque" mode or when motor is stopped

PI Control Kp: Ki:

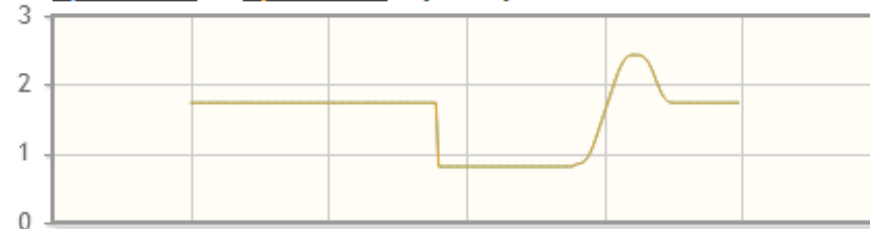
SpinTAC Control Bandwidth:[rad/s]

Torque Mode

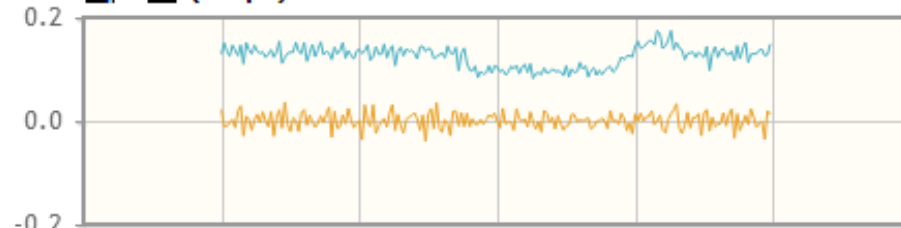
Torque Reference I_q (pu)

-1.00 0.00 1.00

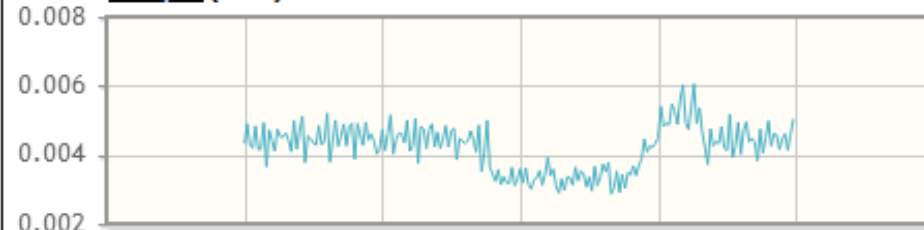
Speed-EST Speed-REF (KRPM)



I_q I_d (Amps)



Torque (N.m)



Estimated Speed (RPM)

1734

Vbus (V)

24

Target Speed (RPM)

1735

Direction



Start-Up

Rs ReCal

☐ Disable
☒ Enable

Offset ReCal

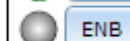
☒ Disable
☐ Enable

Force Angle Start

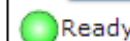
☒ Disable
☐ Enable



ENB Speed+Torque



ENB Torque



Ready, Motor Identified



Motor Running



DRV FAULT STATUS

Start/Stop

Current Control

I_q PI Control

Kp

0 6 12 18 24 30

Ki

0.0 0.1 0.2 0.3 0.4 0.5

Kp

0 6 12 18 24 30

I_d PI Control

Ki

0.0 0.1 0.2 0.3 0.4 0.5



ACIM PowerWarp™ - Minimum Current Use At All Times

STAIRCASE SPIN

	Speed RPM	Acceleration RPM/s	Ramp + Hold Time Seconds
Target 1	1500	220	20
Target 2	2500	220	20
Target 3	3500	262	23
Target 4	4500	275	23

	Speed RPM	Acceleration RPM/s	Ramp + Hold Time Seconds
Target 5	5500	281	31
Target 6	6500	262	23
Target 7	7000	240	25
Deceleration to 0 RPM		50	

Speed Loop Tuning

To Enable SpinTAC, go to "SpinTAC 1" tab.
Switching is allowed only in "Speed+Torque"
mode or when motor is stopped

PI Control

Kp: 8

ENB

Ki: 0.08

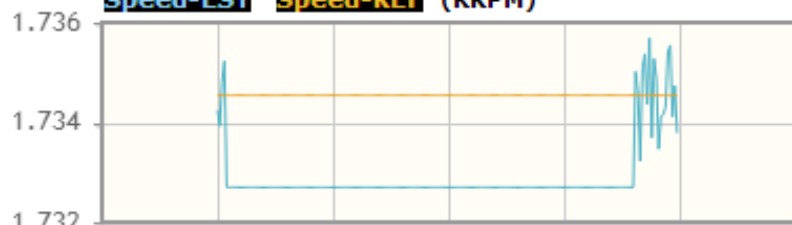
SpinTAC
Control

Bandwidth:[rad/s]

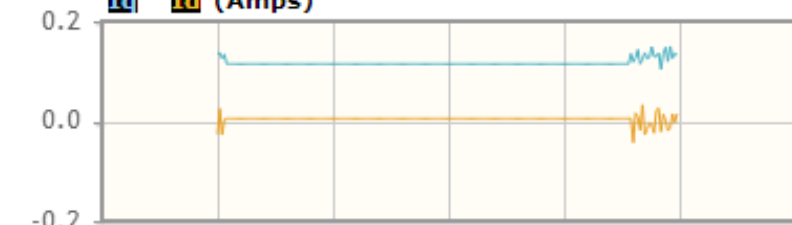
ENB

39.94

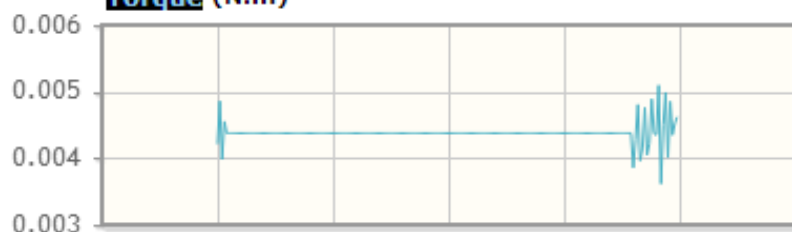
Speed-EST Speed-REF (KRPM)



Iq Id (Amps)



Torque (N.m)

Estimated Speed
(RPM)Target Speed
(RPM)Vbus (V)
24

Direction



Start-Up

0.3919

Rs ReCal

☐ Disable
☒ Enable

Offset ReCal

☒ Disable
☐ EnableForce Angle
Start☒ Disable
☐ Enable

AGITATE / SPEED REVERSAL

RPM cw/ccw

580

Acceleration
RPM/s

580

Ramp + Hold Time
Seconds

5

Deceleration
RPM/s

580

Coast Time
SecondsRamp Time
Seconds

2

- ☐ ENB Staircase Spin
- ☐ ENB Agitate / Speed Reversal
- ☒ Ready, Motor Identified
- ☐ Motor Running
- ☒ DRV FAULT STATUS

Start/Stop

Field Control

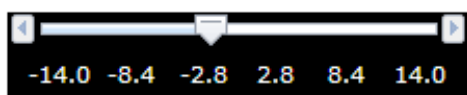
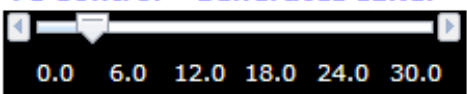
Max Negative Id Allowed (A)



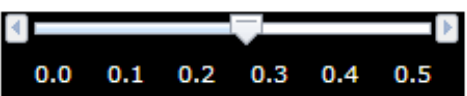
Auto Field Weakening

Manual Field Weakening (-) or Boosting(+)

Id Reference (A)

**PI Control - Generates IdRef**

Kp



Ki

SVM Modulation

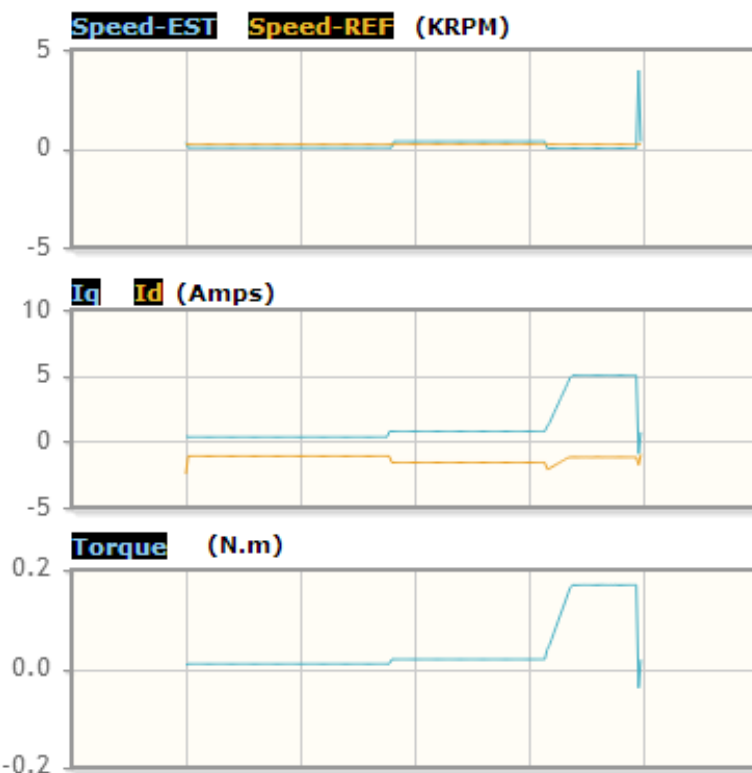
[Max Duty Cycle, 1.15 default]

0.00 to 1.00: Perfect Sinusoidal

1.00 to 1.15: Quasi-Sinusoidal

1.15 to 2.00: Trapezoidal

For use by EXPERTS only in conjunction with other tabs & ~~WARN~~ WARNING Damage may be done to your motor or power stage **WARNING**



Estimated Speed (RPM)

0

Vbus (V)

24

Target Speed (RPM)

0

Direction



Motor Identified



Motor Running

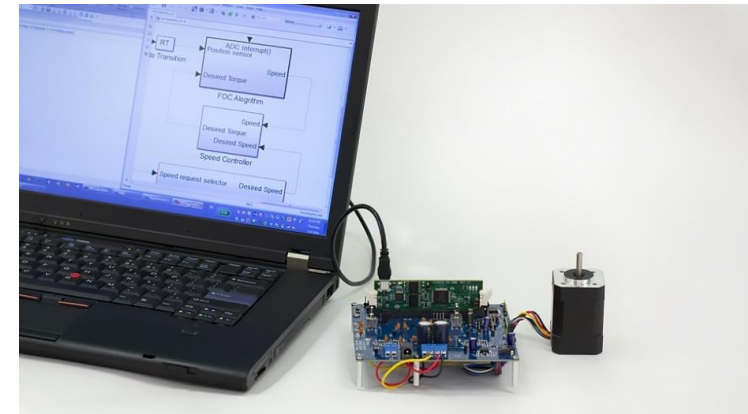
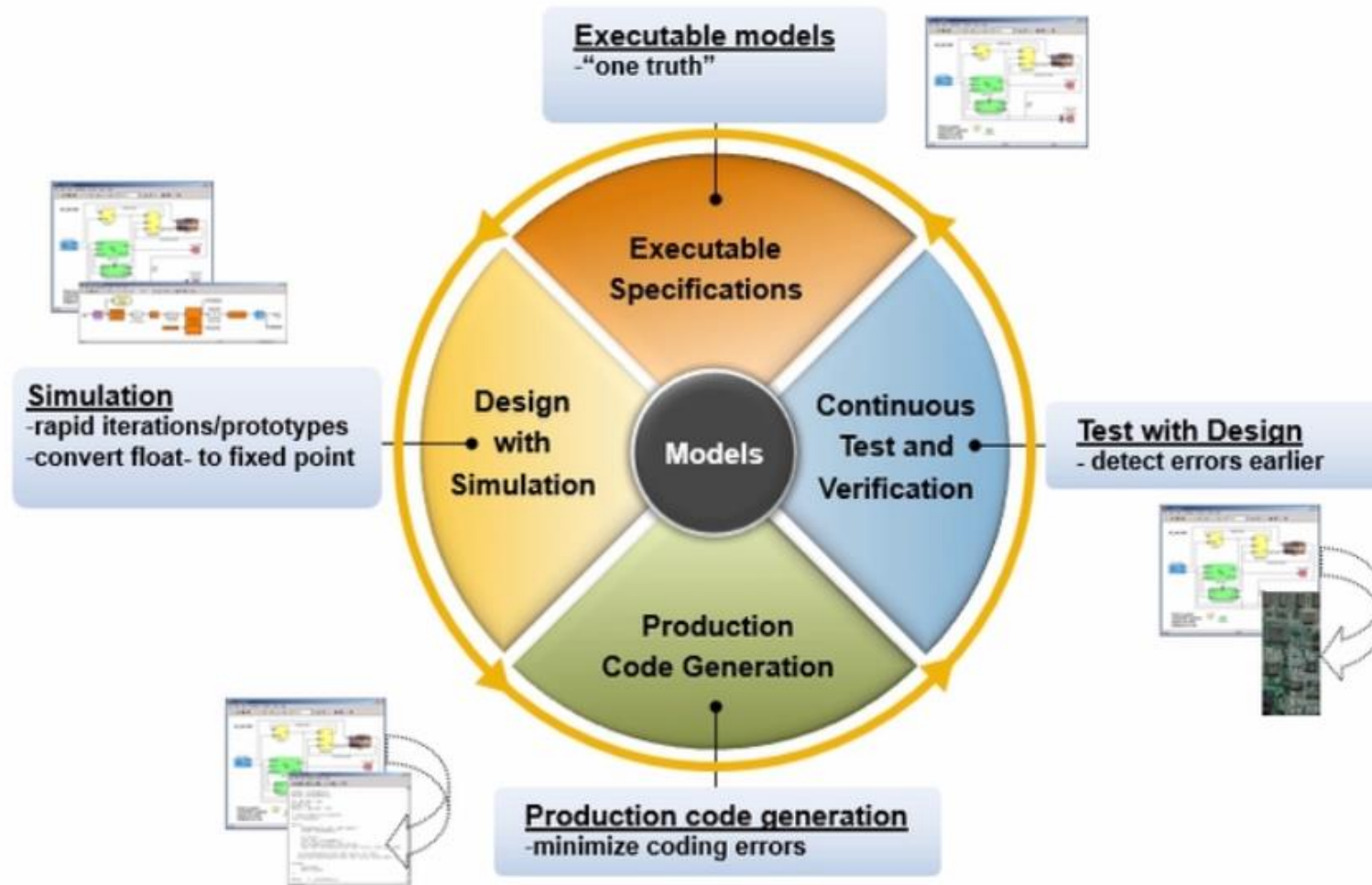


DRV Fault Status

Start/Stop

Model Based Design – Matlab/Simulink

Model-Based Design with Production Code Generation



Model Based Design – Matlab/Simulink

C/C++ Coders

MATLAB Coder - Code from MATLAB

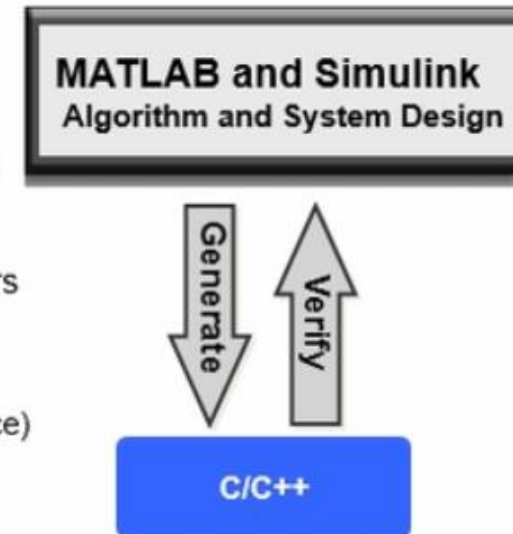
- Portable code for numerical algorithms
- Desktop applications (standalone, library)

Simulink Coder - Code from Simulink

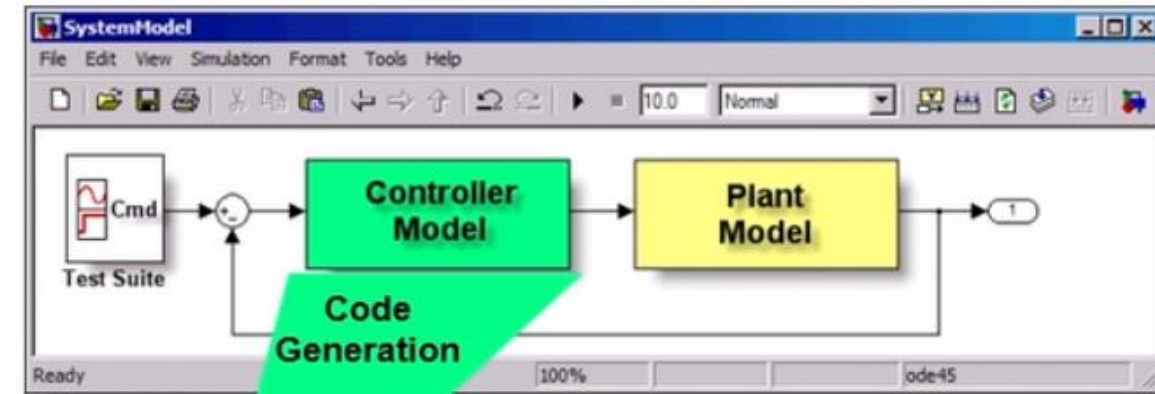
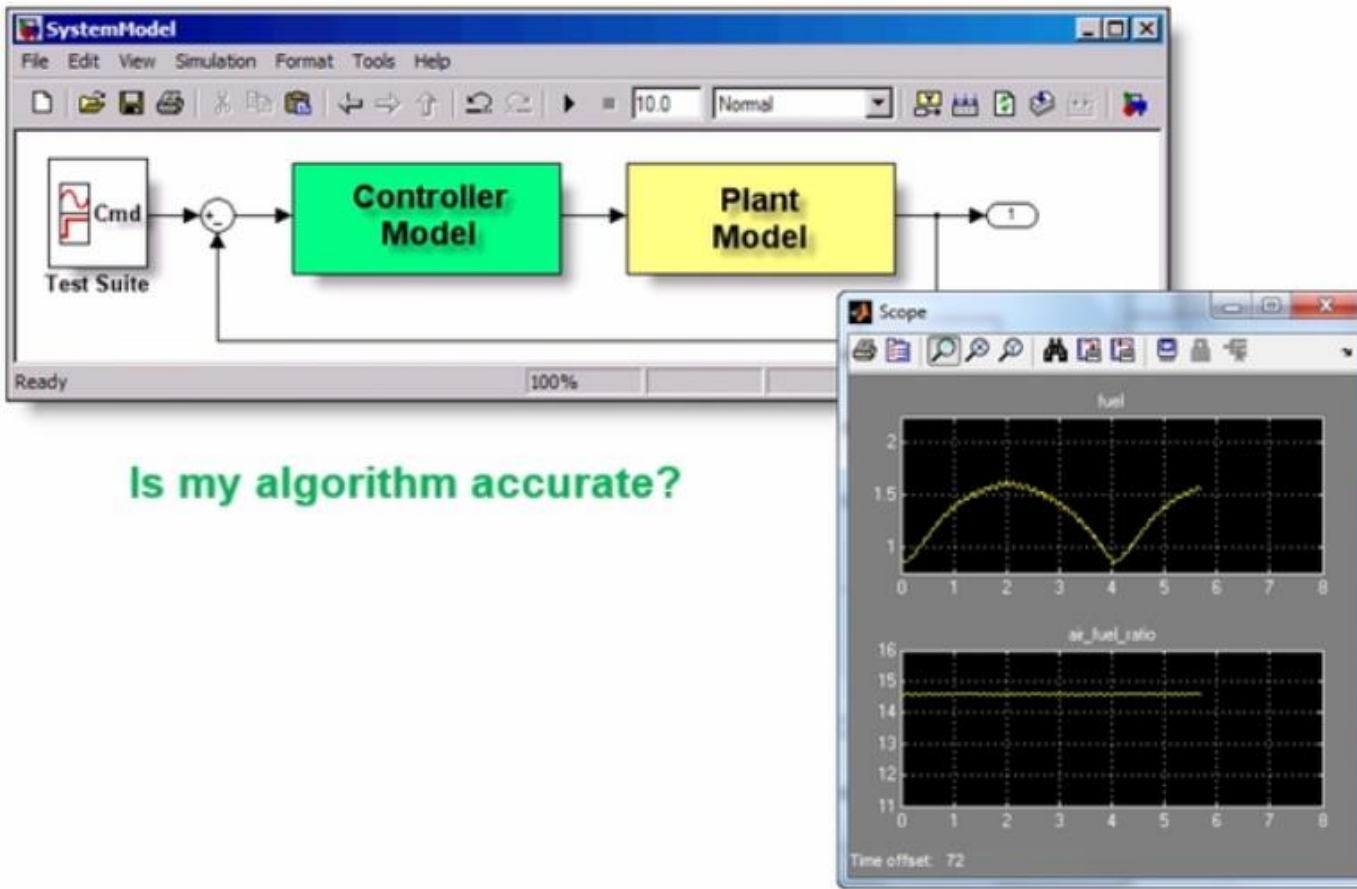
- Portable code for algorithms plus real-time framework
- Real-time machines for RP/HIL (e.g., Simulink Real-Time)

Embedded Coder – Production code

- Extends ML Coder and SL Coder for embedded processors
- MCUs and DSPs (from 8-bit devices to multicore SoCs)
 - Code optimization (portable code and processor-specific)
 - Code verification (software- and processor-in-the-loop, trace)
 - Code profiling (tasks and functions)
 - Code customization (data, functions, files)
 - Embedded targets (board initialization, I/O blocks, scheduler)
 - Certification (ISO-26262, IEC 61508, etc.)



Model Based Design – Matlab/Simulink



Embedded Processor or ECU

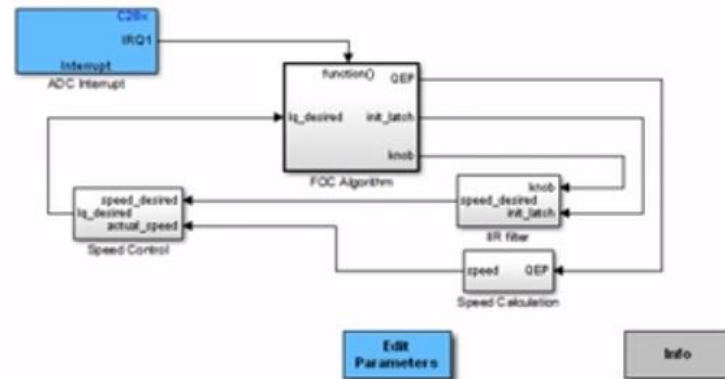
Design with Simulation

Embedded Coder support for TI C2000 Motor Control kits.

- TI F28027 Launchpad + DRV8301 ([FX + Video](#))
- TI F28069 Launchpad + 2 x DRV8301 ([FX + Video](#))
- ControlCard + DRV8312 ([Shipping example](#))
- DM550 + eZdsp ([Shipping example](#))
- High voltage motor control kit (available on demand)

Permanent Magnet Synchronous Motor Field-Oriented Control

Note: This demo requires a DMC550 controller and a PMS motor



Copyright 2006-2015 The MathWorks, Inc.



Design with Simulation

Model : Double

Embedded Processor : Single

Bu ayarlar Matlab/Simulink'te tasarlanan sistemin donanım üzerinde düzgün çalışmasını sağlıyor. Bu uyumsuzluklar

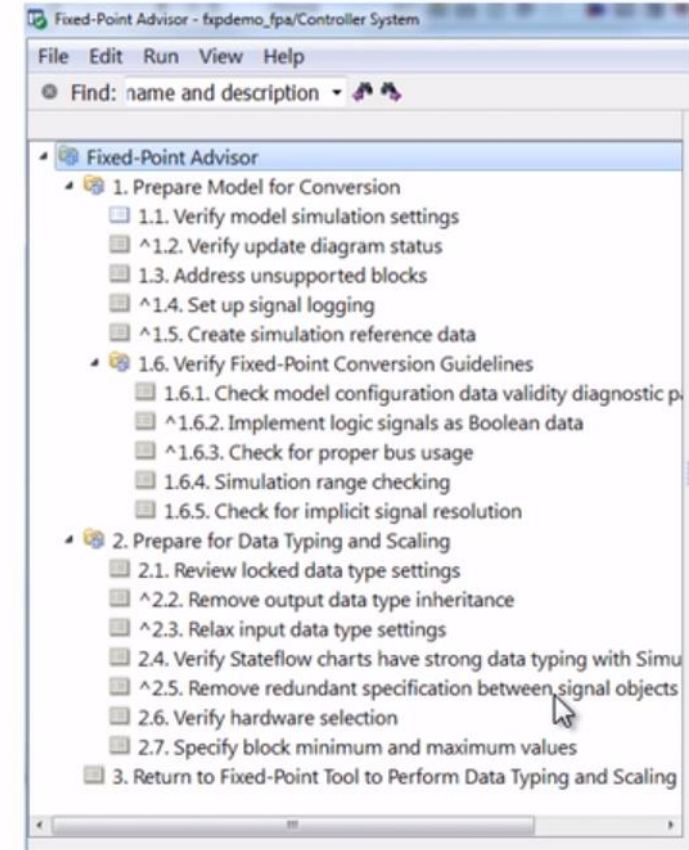
- Integer types
- Floating points
- Input/Output Ranges
- Execution ranges

Olabilir.

Float- to Fixed-Point Conversion

- Overflow/underflow
- Code optimizations
- Simulation ranges
- Derived ranges
 - Design-range scaling

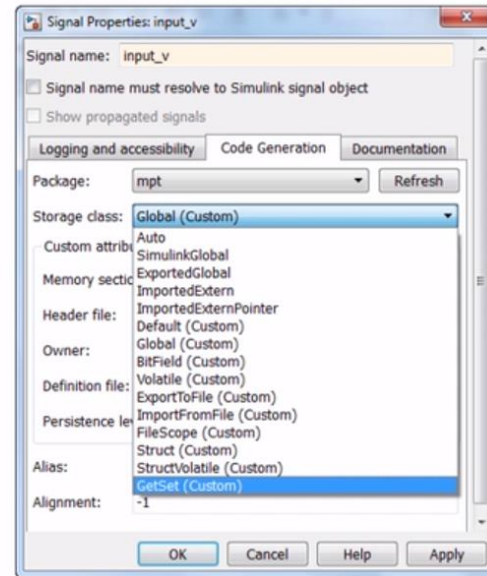
>>fxpdemo_fpa



Design with Simulation

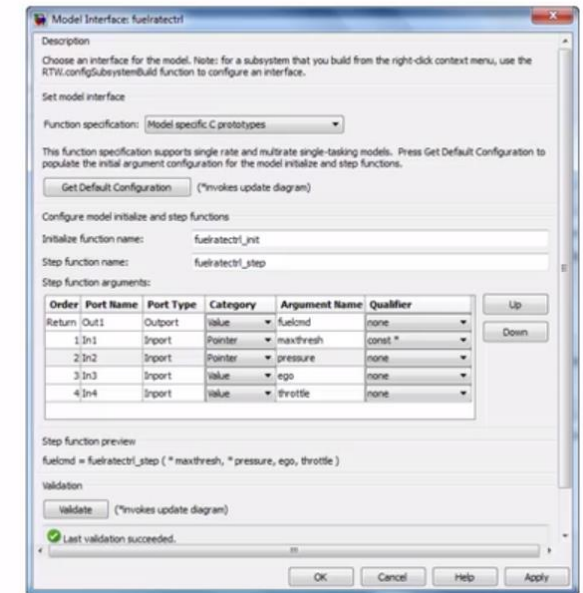
Data Specification

- Name
- Storage class
- Alias (typedef)
- Comments



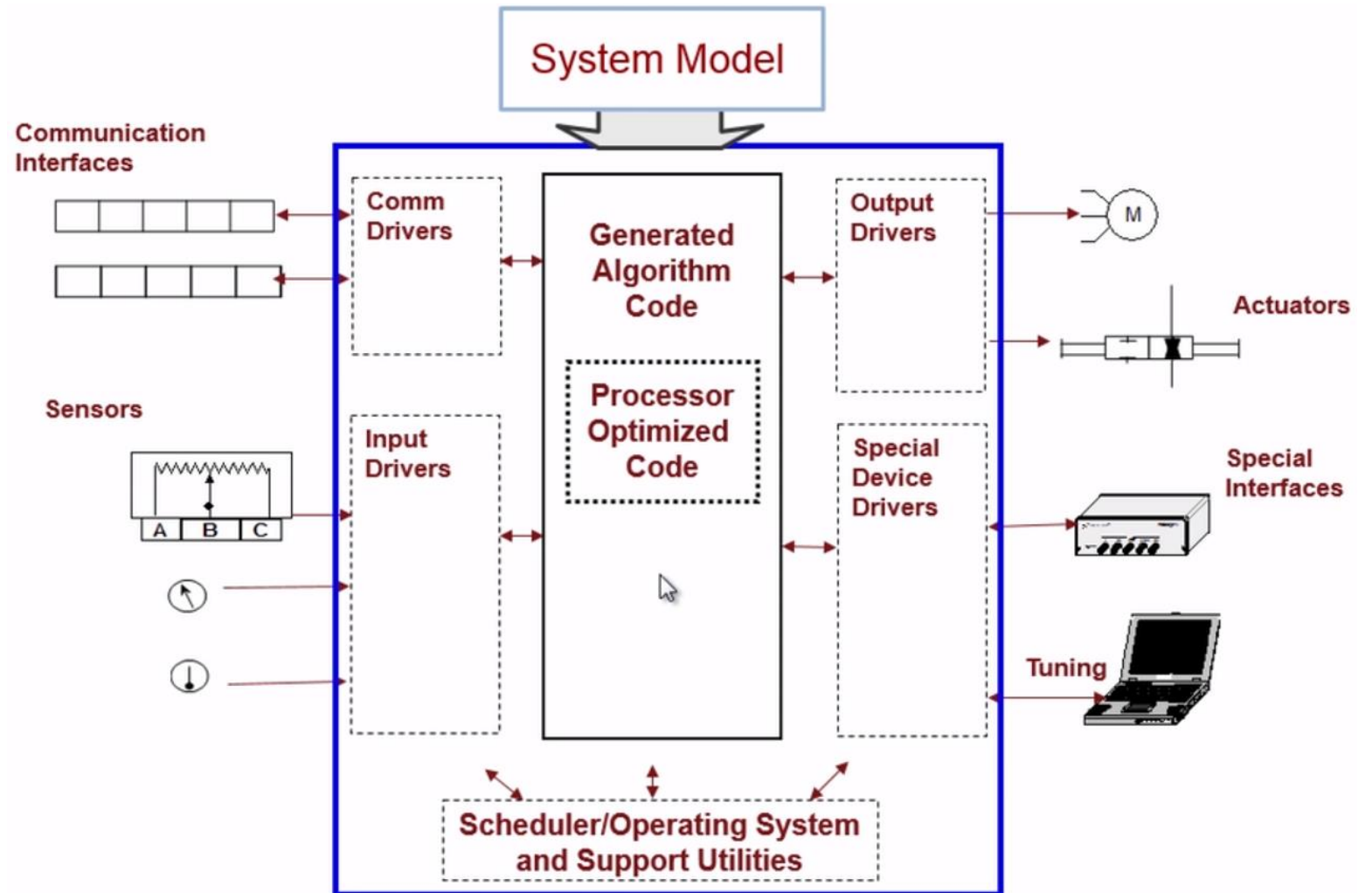
Function Interface Specification

- Function Name
- Argument Name
- Pass by value
- Pass by reference
- Qualifier



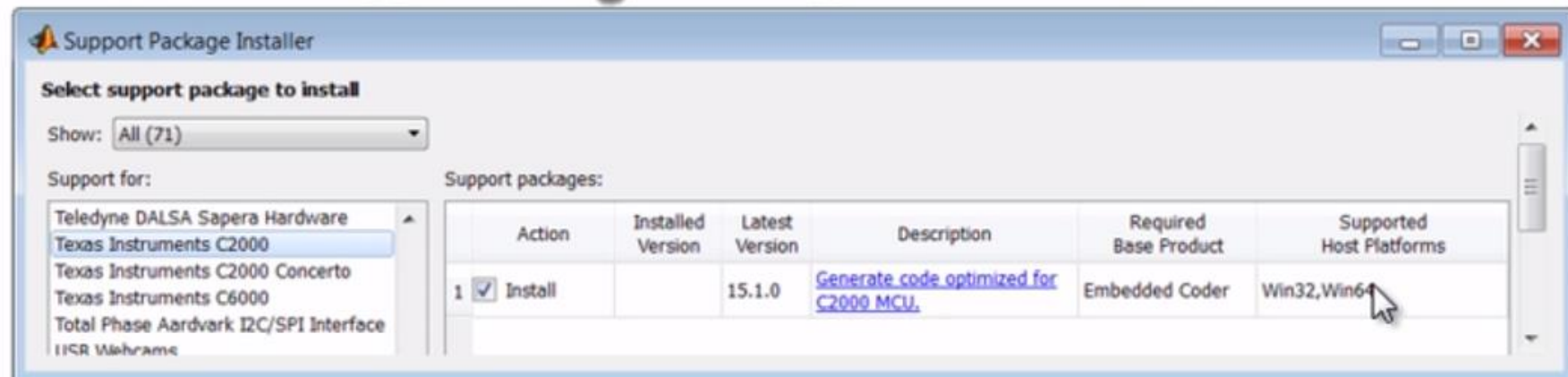
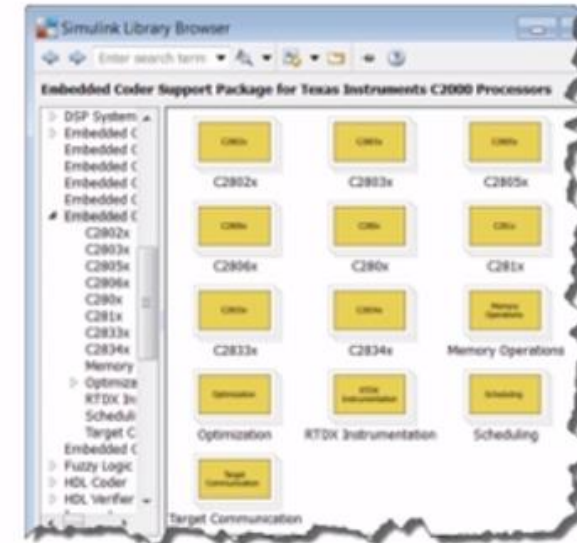
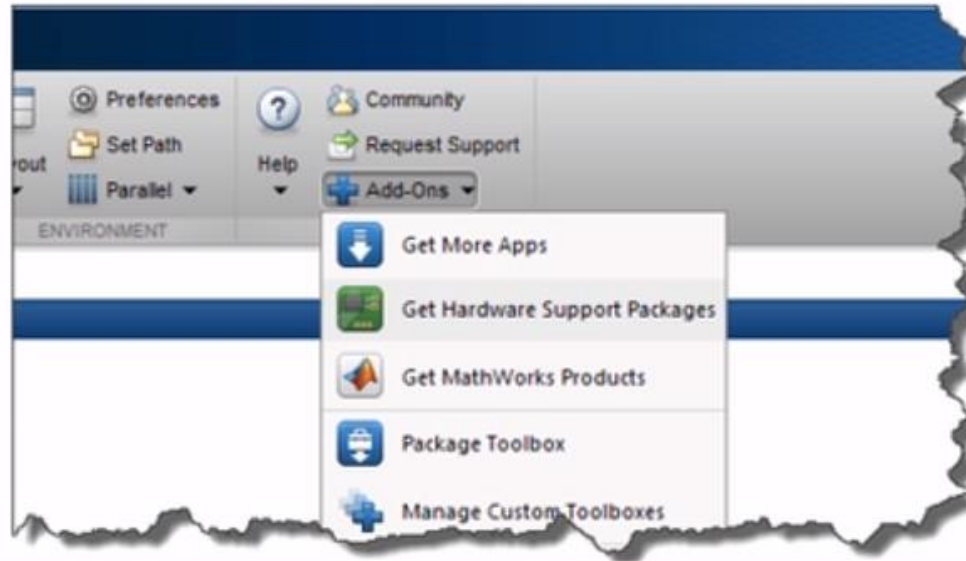
Design with Simulation

- Algorithm code
- Processor Optimization



Design with Simulation

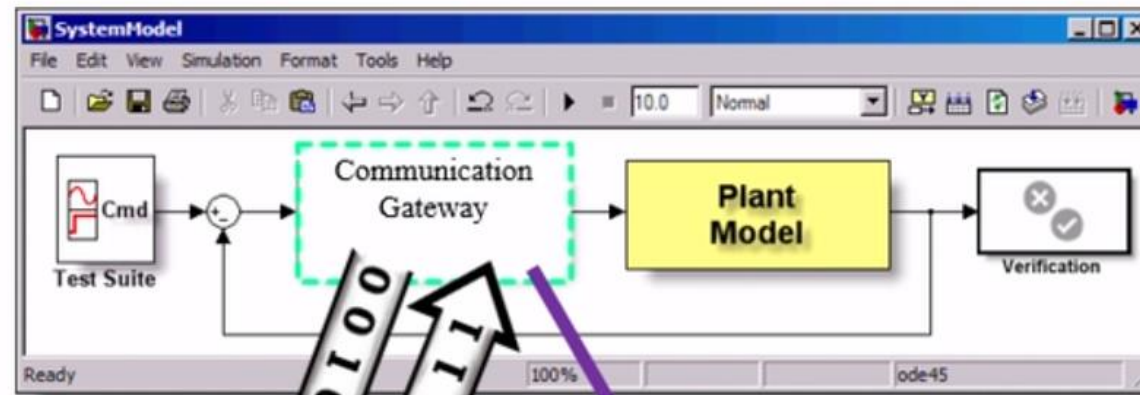
Embedded Coder Hardware Support Packages



Design with Simulation

Processor-in-the-Loop (PIL)

Verify algorithms



Non-Real-Time
Synchronization
with Host at Each Time Step

Supported devices

- Piccolo F2802x/3x/5x/6x (07x in progress)
- Delfino F2833x/32x (F2837x in progress)
- Fixed-point F280x/1x
- Concerto (F28M35/M36)



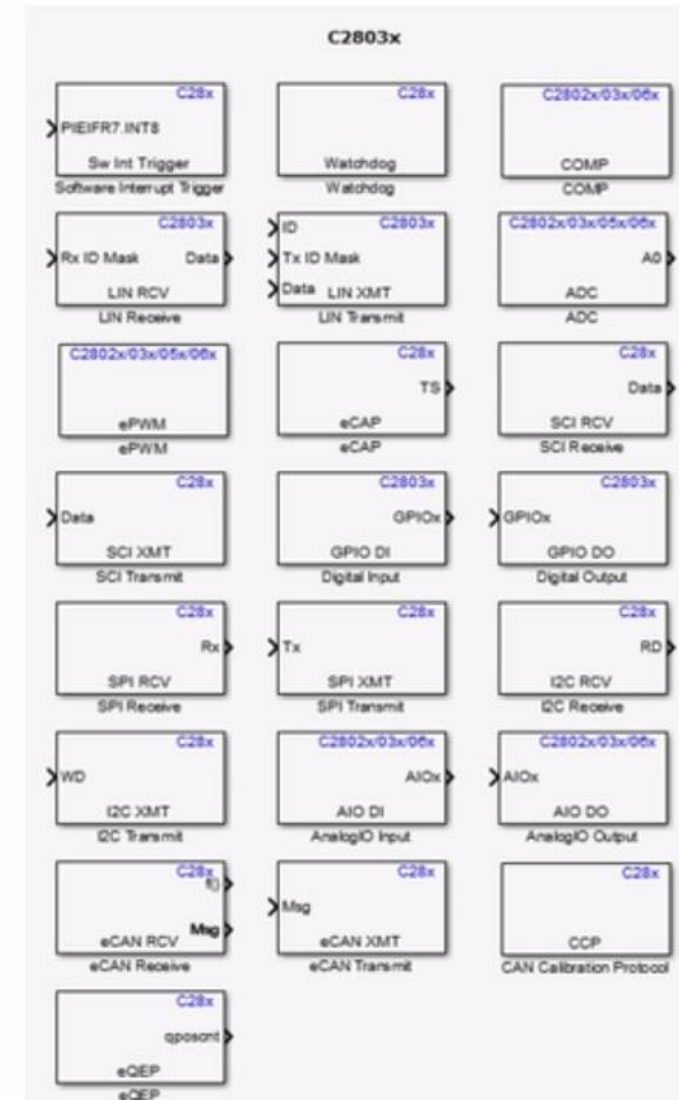
Execution History

- Results comparison (sim vs. tgt)
- Code coverage
- Execution timing

Design with Simulation

Supported TI C2000 drivers

- ADC, AIO, Comparator,
 - GPIO, eQEP, ePWM, eCAP,
 - eCAN, I2C, SCI, SPI, LIN
 - Watchdog, DMA.
-
- Motor control position sensing
 - Optical encoder (using eQEP)
 - Hall sensors (using eCAP)
 - Sensorless (Using SMO)





Summary

Code generation for model "c28069pmsmfocdual"

Configuration settings at the time of code generation: [click to open](#)

Code generation objectives: **Unspecified**

Validation result: Not run

Generated Code

[-] Main file

ert_main.c

[-] Model files

[c28069pmsmfocdual.c](#)
[c28069pmsmfocdual.h](#)
[c28069pmsmfocdual_priv](#)
[c28069pmsmfocdual_type](#)

[-] Data files

c28069pmsmfocdual_data

```

Diagnostic Viewer
c28069pmsmfoc...
output_file="ert_main.obj"
"C:\Work\Devel\F28069_launchpad\c28069pmsmfocdual_ert_rt\ert_main.c"
"C:\Work\Devel\F28069_launchpad\c28069pmsmfocdual_ert_rt\ert_main.c", line 106:
warning: variable "modelBaseRate" was declared but never referenced
"C:\Work\Devel\F28069_launchpad\c28069pmsmfocdual_ert_rt\ert_main.c", line 107:
warning: variable "systemClock" was declared but never referenced
"### Creating standalone executable "../c28069pmsmfocdual.out" ..."
C:\PROGRA~2\TEXAS~1\F2800~1.9\bin\cl2000 -z -IC:\PROGRA~2\TEXAS~1\F2800~1.9\lib --
stack_size=1024 --warn_sections --reread_libs --rom_model -m"c28069pmsmfocdual.map" -
1"rts2800_fpu32.lib" --define BOOT_FROM_FLASH=1 --output_file="../c28069pmsmfocdual.out
MW_c28xx_board.obj DSP28xx_SciUtil.obj MW_c28xx_adc.obj MW_c28xx_csl.obj
MW_c28xx_pwm.obj c28069pmsmfocdual.obj c28069pmsmfocdual_data.obj
c2806xBoard_Realtime_Support.obj F2806x_CpuTimers.obj F2806x_DefaultIsr.obj
F2806x_GlobalVariableDefs.obj F2806x_PieCtrl.obj F2806x_PieVect.obj F2806x_SysCtrl.obj
F2806x_usdDelay.obj F2806x_CodeStartBranch.obj F2806x_Dma.obj F2806x_Adc.obj
c2806xScheduler_ADINT1.obj ert_main.obj
C:\MATLAB\SupportPackages\R2014b\texasinstrumentsc2000\toolbox\target\supportpackages\
tic2000\src\c28069m.cmd
C:\MATLAB\SupportPackages\R2014b\texasinstrumentsc2000\toolbox\target\supportpackages\
tic2000\rtlib\IQmath_fpu32.lib
C:\PROGRA~2\MATLAB\R2014b\toolbox\idmlink\extensions\ticcs\c2000\c2806xPeripherals.cmd

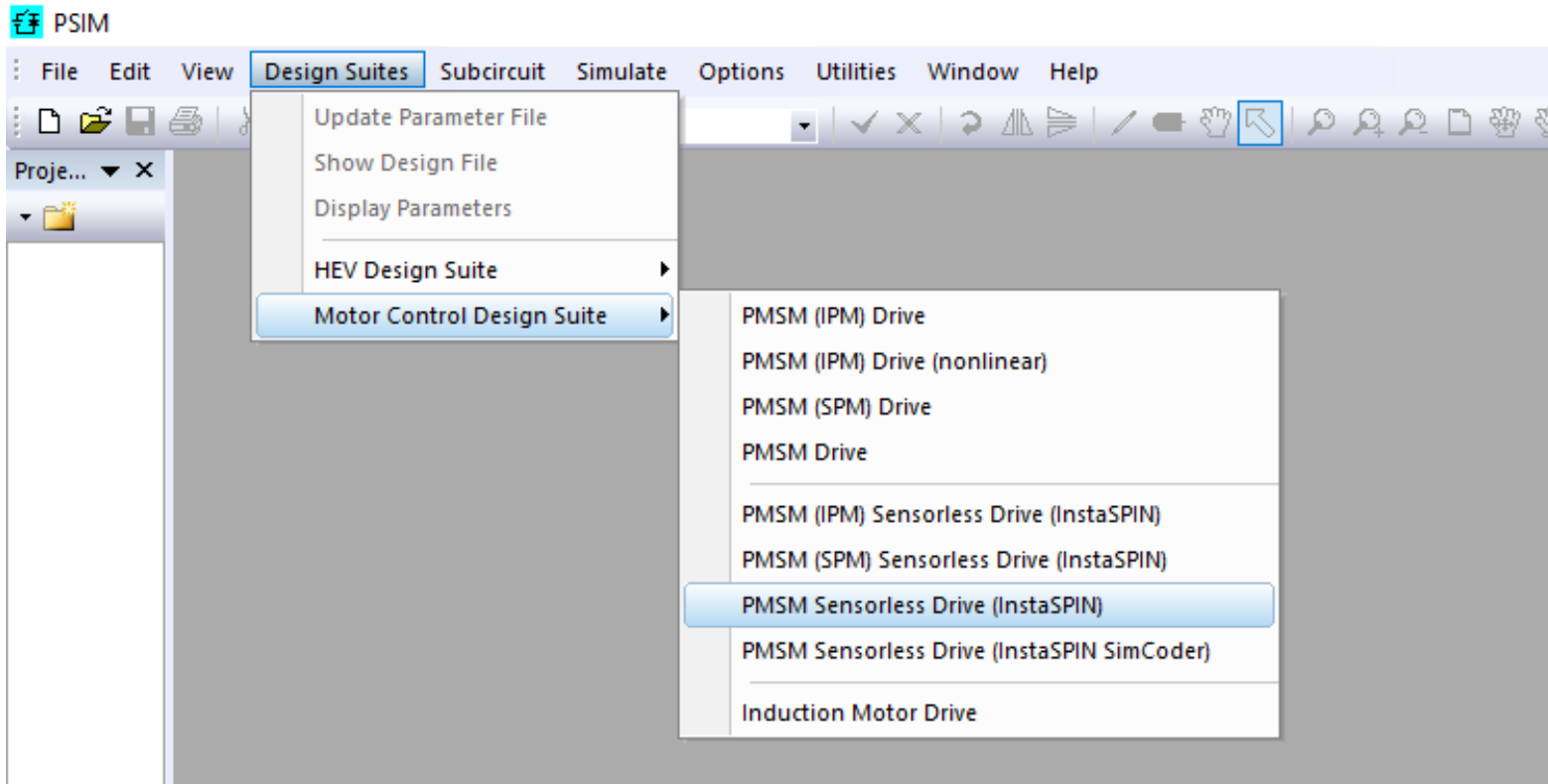
<Linking>
warning: build attribute vendor section TI missing in
"C:\MATLAB\SupportPackages\R2014b\texasinstrumentsc2000\toolbox\target\suppo
rtpackages\tic2000\rtlib\IQmath_fpu32.lib\IQ17atan2PU.obj": compatibility
cannot be determined
warning: build attribute vendor section TI missing in
"C:\MATLAB\SupportPackages\R2014b\texasinstrumentsc2000\toolbox\target\suppo
rtpackages\tic2000\rtlib\IQmath_fpu32.lib\IQ17cosPU.obj": compatibility
cannot be determined
warning: build attribute vendor section TI missing in
"C:\MATLAB\SupportPackages\R2014b\texasinstrumentsc2000\toolbox\target\suppo
rtpackages\tic2000\rtlib\IQmath_fpu32.lib\IQ17sinPU.obj": compatibility
cannot be determined
warning: build attribute vendor section TI missing in
"C:\MATLAB\SupportPackages\R2014b\texasinstrumentsc2000\toolbox\target\suppo
rtpackages\tic2000\rtlib\IQmath_fpu32.lib\IQmathTables.obj": compatibility
cannot be determined
"### Created: ../c28069pmsmfocdual.out"
"### Successfully generated all binary outputs."

C:\Work\Devel\F28069_launchpad\c28069pmsmfocdual_ert_rt\exit /B 0
"### Invoking postbuild tool "Download" ..."
C:\ti\ccsv5\ccs_base\SCRIPT-1\bin\dss.bat
C:\PROGRA~2\MATLAB\R2014b\toolbox\idmlink\EXTENS~1\ticcs\ccsdemos\RUNPRO-1.35
C:\MATLAB\SUPPOR~1\R2014b\TEXAS~1\toolbox\target\SUPPOR~1\tic2000\BSCUS~1\..\CC5_CO~
1\F28069~1.CCX ../c28069pmsmfocdual.out

```

Model Based Design - PSIM

- Need PSIM full professional version 10, 11 or 12.



Motor identification

Identify motor parameters and voltage/current offsets with Motor ID code in PSIM

Parameter preparation

Generate InstaSPIN parameters based on motor and hardware setup using InstaSPIN Parameter Editor in PSIM

Controller design

Design current/speed loop controller parameters using Motor Control Design Suite in PSIM

Simulation

Simulate power converter and controller in digital control

Simulation for DSP

Simulate power converter and fixed-point controller with DSP peripheral blocks (ADC, PWM, etc.)

Code generation

Auto generate code for DSP

Upload to DSP



F28069M or other InstaSPIN-enabled DSP hardware



Proj... X

main - PMSM Sensorless... X

Update Parameter File

DC Bus

Vdc 24

Inverter

Rshunt 7m

Kgain_i -10

I_offset 1.65

R1_v 62k

R2_v 4.99k

Motor

J_motor 4.8u

T_shaft 0.05

Load

T_load 0

J_load 4u

Motor Controller

fsam_i 15k

fsam_w 1k

fcr_i 1k

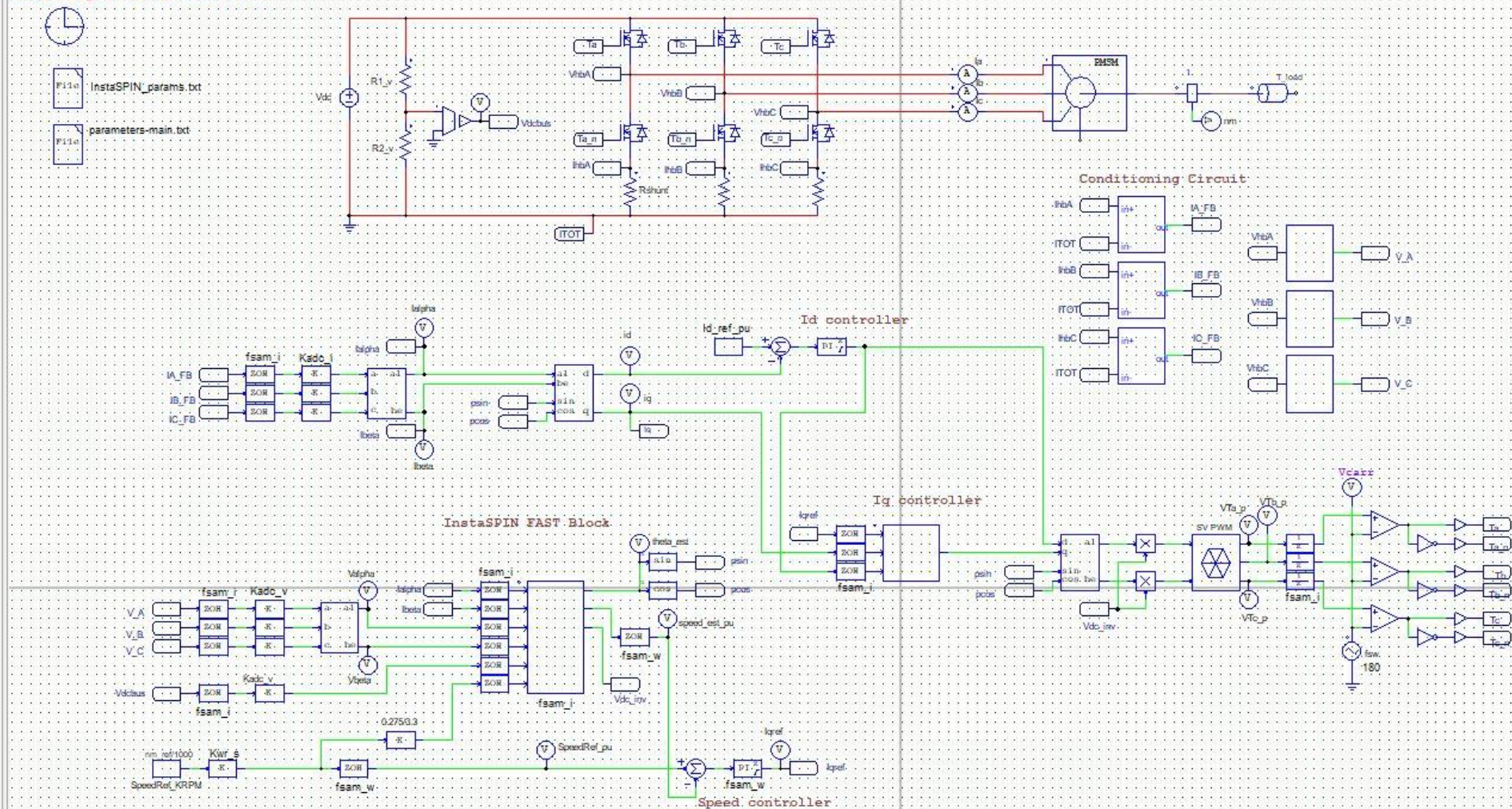
fcr_w 25

Id_ref 0

nm_ref 500

Update Parameter File

Note: Simulating this circuit requires the PLE Module.





Proje... ▾ X

main - PMSM Sensorless.psi

main - PMSM Sensorless...

Update Parameter File

DC Bus

Vdc 24

Inverter

Rshunt 7m

Kgain_i -10

I_offset 1.65

R1_v 62k

R2_v 4.99k

Ia_offset 0.997648

Ib_offset 1.00098

Ic_offset 0.998795

Va_offset 0.49609

Vb_offset 0.4954

Vc_offset 0.4964

Motor

J_motor 4.8u

T_shaft 0.05

Load

T_load 0

J_load 4u

Motor Controller

fsam_i 15k

fsam_w 1k

fcr_i 1k

fcr_w 25

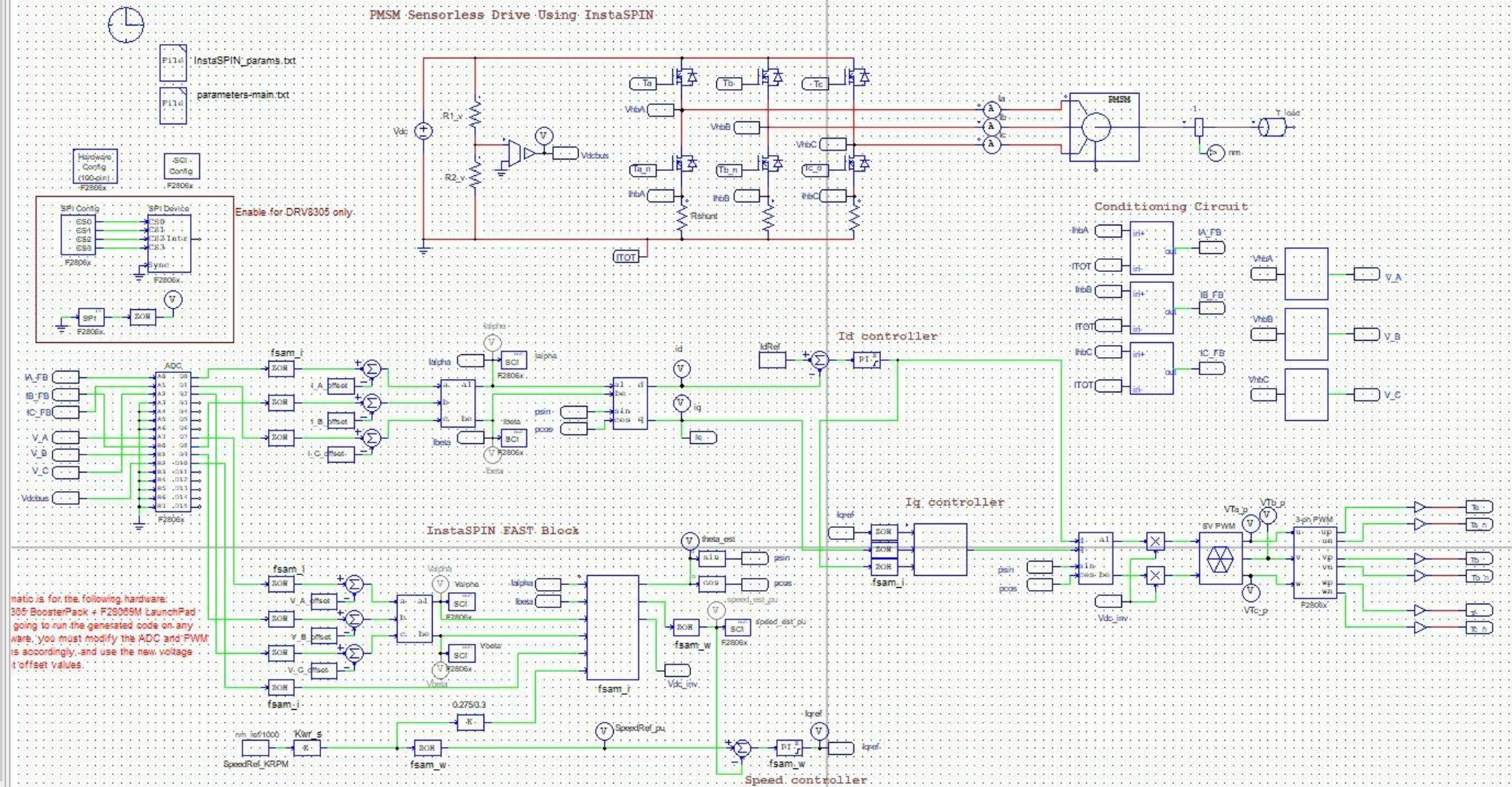
Id_ref 0

nm_ref 500

Update Parameter File

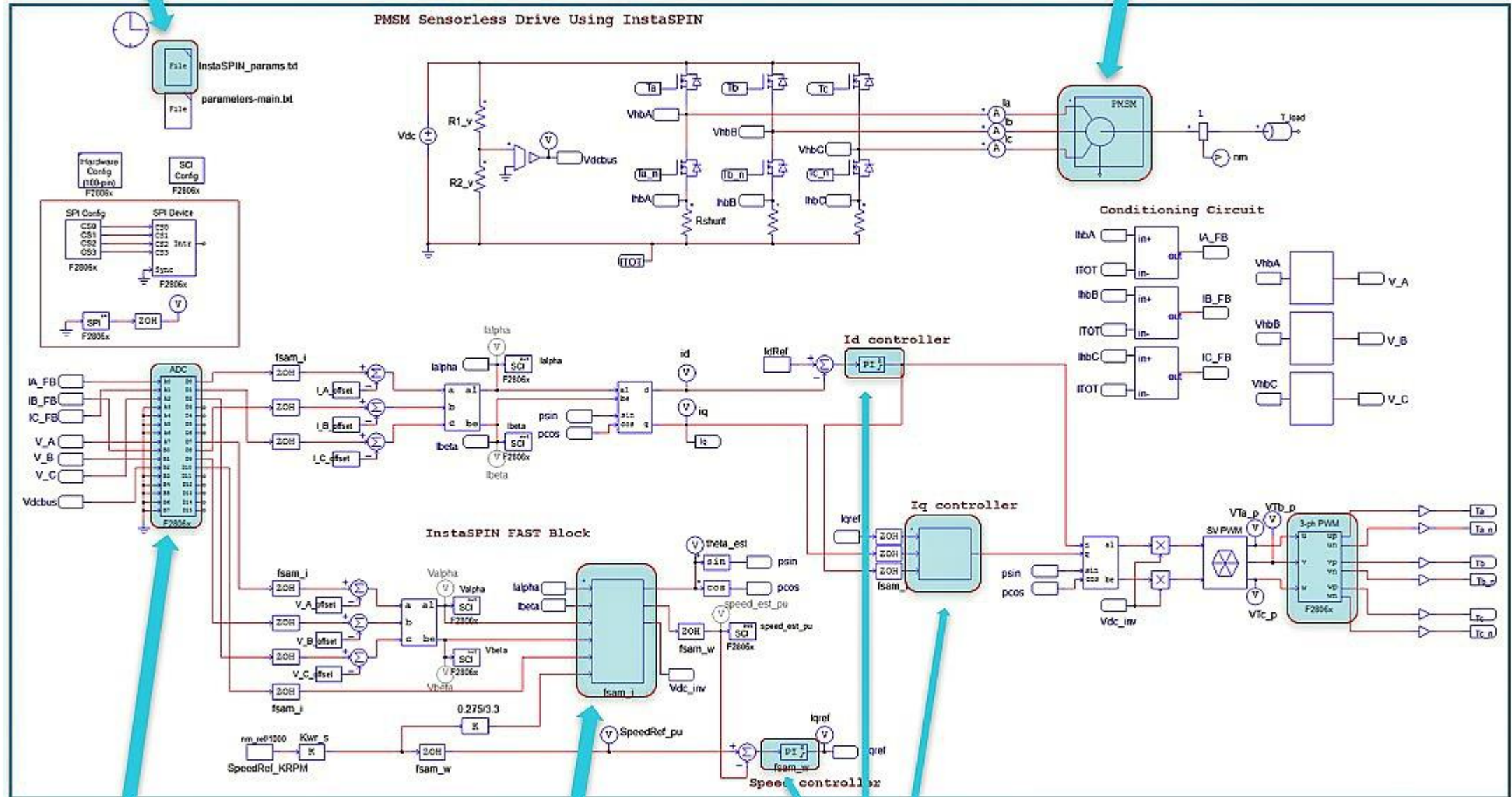
PIL: SimCoder: F2806x Target

PMSM Sensorless Drive Using InstaSPIN



Parameter File Preparation

Motor Identification



Auto Code Generation

Simulation

Controller Design



Project View

DRV8312 PMSM InstaSPIN Lab11 (F28069)

```
//*****  
// This code is created by SimCoder Version 12.0.3 for F2806x Hardware Target  
//  
// SimCoder is copyright by Powersim Inc., 2009-2019  
//  
// Date: January 27, 2020 17:17:31  
//*****  
#define GLOBAL_Q 24  
long GlobalQ = GLOBAL_Q;    // Used for legacy GEL & Graph Debug.  
#include "IQmathLib.h"  
#include "PS_bios.h"  
#define GetCurTime() PS_GetSysTimer()  
#define PWM_IN_CHECK    // To lower PWM value setting time, comment out this line if PWM duty cycle values are strictly limited in the range.  
  
#include "include/clarke.h"  
#include "include/est.h"  
#include "include/ctrl_obj.h"  
#include "include/ctrl_states.h"  
#include "include/est_Flux_states.h"  
#include "include/est_Ls_states.h"  
#include "include/est_Rr_states.h"  
#include "include/est_Rs_states.h"  
#include "include/est_states.h"  
#include "include/est_Traj_states.h"  
#include "include/filter_fo.h"  
#include "include/ipark.h"  
#include "include/math.h"  
#include "include/motor.h"  
#include "include/offset.h"  
#include "include/park.h"  
#include "include/pid.h"  
#include "include/stdint.h"  
#include "include/svgen.h"  
#include "include/traj.h"  
#include "include/types.h"  
#include "include/userParams.h"  
#include "include/ctrl.h"  
#include "user.h"  
  
interrupt void Task0;  
interrupt void Task_10;  
void TaskS20;  
<
```

Simulation Message

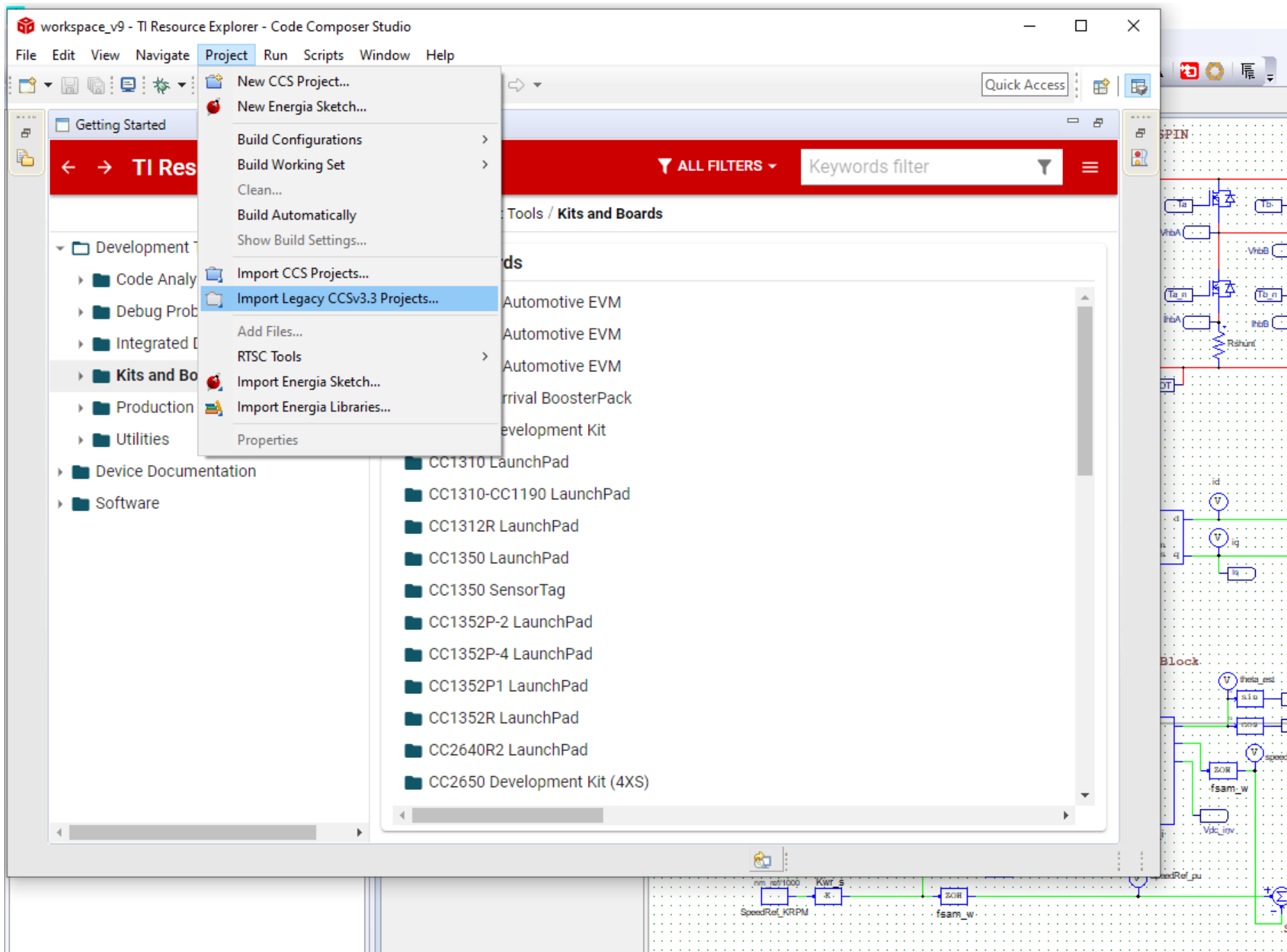
Project View Library Browser



Ready

Type here to search





workspace_v9 - DRV8312_PMSM_InstaSPIN_Lab11_F28069/DRV8312_PMSM_InstaSPIN_Lab11_F28069_c - Code Composer Studio

File Edit View Project Tools Run Scripts Window Help

Debug

DRV8312_PMSM_InstaSPIN_Lab11_F28069_ [Code Composer Studio - Device Debugging]

Texas Instruments XDS100v2 USB Debug Probe_0/C28xx (Suspended)

Task() at DRV8312_PMSM_InstaSPIN_Lab11_F28069_c:1,024 0x008C7A

main() at DRV8312_PMSM_InstaSPIN_Lab11_F28069_c:1,341 0x008C3D

updateGlobalVariables(struct EST_Obj_ *) at user.c:149 0x00A343

_ProcSciWaitStart() at DRV8312_PMSM_InstaSPIN_Lab11_F28069_c:207 0x008809

0x3A8A08 (no symbols are defined)

Texas Instruments XDS100v2 USB Debug Probe_0/CLA_0 (Disconnected : Unknown)

Variables Expressions Registers

Name	Type	Value	Location
fADC1_1	long	-228724238	0x0000009E@Data
fADC1_10	long	531299264	0x000000B6@Data
fADC1_12	long	-935963584	0x000000B2@Data
fADC1_13	long	66600	0x000000A2@Data
fADC1_15	long	-285126	0x000000AA@Data
fADC1_5	long	-285126	0x000000A6@Data
fADC1_7	long	-931939200	0x000000AE@Data
fC10	long	79936	0x0000008E@Data
fC11	long	79966	0x00000090@Data

DRV8312_PMSM_InstaSPIN_Lab11_F28069_c

```
1/*****
2// This code is created by SimCoder Version 12.0.3 for F2806x Hardware Target
3//
4// SimCoder is copyright by Powersim Inc., 2009-2019
5//
6// Date: January 27, 2020 17:17:31
7 *****/
8#define GLOBAL_Q 24
9long GlobalQ = GLOBAL_Q; // Used for legacy GEL & Graph Debug.
10#include "IQmathLib.h"
11#include "PS_bios.h"
12#define GetCurTime() PS_GetSysTimer()
13#define PWM_IN_CHECK // To lower PWM value setting time, comment out this line if PWM duty cycle values are strictly limited in the range.
14
15
16
17#include "include/clarke.h"
18#include "include/est.h"
19#include "include/ctrl_obj.h"
20#include "include/ctrl_states.h"
21#include "include/est_Flux_states.h"
22#include "include/est_Ls_states.h"
23#include "include/est_Rr_states.h"
24#include "include/est_Rs_states.h"
25#include "include/est_states.h"
26#include "include/est_Traj_states.h"
27#include "include/filter_fo.h"
28#include "include/ipark.h"
29#include "include/math.h"
```

Console

CDT Build Console [DRV8312_PMSM_InstaSPIN_Lab11_F28069_]

**** Build of configuration 1_RamDebug for project DRV8312_PMSM_InstaSPIN_Lab11_F28069_ ****

"C:\ti\ccs930\ccs\utils\bin\gmake" -k -j 12 all -O

gmake[1]: 'C:/Powersim/PSIM12.0.3_TrialX64/examples/SimCoder/F2806x Target/TI DRV8312 PMSM InstaSPIN Lab11/DRV8312 PMSM InstaSPIN Lab11 (F28069) (C code)/DRV8312_PMSM_InstaSPIN_Lab11_F28069_../RamDebug/DRV8312_PMSM_InstaSPIN_Lab11_F28069_.out' is up to date.

**** Build Finished ****

Writable Smart Insert 1024 : 1

Thanks for your attention.

